PHEMTO: Simulations of Focal Plane

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1 PHEMTO

PHEMTO is a M8 ESA mission proposal that aims to be sensitive to photons from 1 to 400 keV. The observatory aims to improve the sensitivity in the hard X-ray band by several orders of magnitude. Also it will provide tens of arcsec (10 arcsec) imaging capabilities that will allow spectroscopic, polarimetric and imaging studies of the non-thermal universe. PHEMTO will employ a combined X-ray and Gamma-ray optical system to focus the incoming light into the detector system. This configuration allows a very large reduction of cosmic X-ray background on the detectors and therefore very beneficial for both spectral sensitivity and polarimetric sensitivity.

The detector system will be composed of a two layer detector configuration. The top layer, Low Energy Detector (LED), will be a Si detector and the bottom layer a CdTe detector, High Energy Detector (HED). The LED will be used to detect low energy photons, from 1 to 20 keV and to act as a scattering layer for Compton events between the LED and HED. The HED will cover the energy range 4 to 400 keV and will allow to perform polarimetry from 15 to 400 keV.

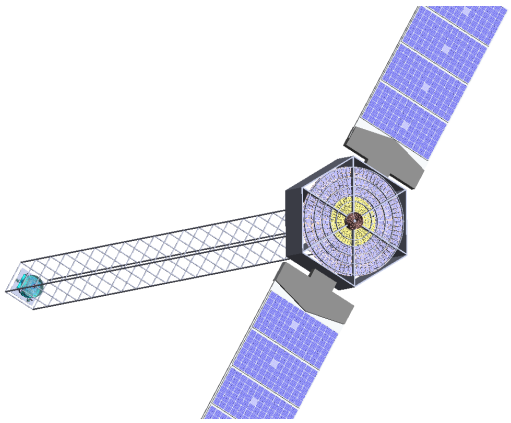


Figure 1: CAD model of PHEMTO. The central red part is the mirror, the cyan part is the Lens part made with Silicon tiles and the yellow part is the ones made with Germanium tiles. We see the detector payload in blue at the end of the boom 20 m away from the optics system. The platform behind is not shown.

To reduce the cosmic X-ray background component for lower energies on the PSF area of the detector system, we will use a 0.13 cm thick lead (Pb) cylindrical collimator positioned 5 cm above the LED. Taking into consideration the lens size, viewed at the focal length of 20 m we have determined its aperture angle to be ¡ 5°, implying a collimator length around 100 cm and an inner radius of 7.6 cm. This configuration will allow ¿99 % X-ray diffuse background suppression at the PSF area for energies until ≈150 keV.

1.1 Compton Polarimetry

Polarimetry in the energy range of a few keV up to few MeV can be achieved by reconstructing the azimuthal direction distribution of the scattered photon. For a polarized photon beam, the azimuthal distribution of the scattered photon is not isotropic but it is related to the electric vector of the incoming photons. This relation is described by the Klein-Nishina differential cross-section for a free electron at rest (we can assume this since energy of incoming photon is ≫ energy electrons bound to the atoms),

$$\frac{d\sigma}{d\Omega} = \frac{r_0^2\epsilon^2}{2}\left(\frac{E'}{E_0} + \frac{E_0}{E'} - 2sin^2\theta cos^2\phi\right) \tag{1}$$

where ϕ is the azimuthal scattering angle between the polarization unit vector of the incident photon and the scattering plane, r_0 is the classical electron radius, E_0 the energy of the incoming photon, E' the energy of the scattered photon and θ the polar angle of the scattered photon [lei]. We can see from equation 1 that for a fixed θ angle the azimuthal direction distribution of the scattered photon follows a cos^2 modulation. In addition we see that when ϕ is 90° to the incoming electric field direction the cross section is at a minimum. This asymmetry allows the measurement of the incoming light polarization angle, where the incoming light electric field direction is perpendicular to the maximum of the azimuthal direction distribution of the scattered photons.

To assess the relative effectiveness of any detector arrangement as a polarimeter it is typically used the polarimetric modulation factor Q metric. It is defined as,

$$Q = \frac{N_\perp - N_\parallel}{N_\perp + N_\parallel} \tag{2}$$

where $N_\perp$ and $N_\parallel$ are the counts in orthogonal directions in the XY plane of the detector. It can also be shown as a function of Klein-Nishina cross section,

$$Q = \frac{d\sigma(\phi = 90°) - d\sigma(\phi = 0°)}{d\sigma(\phi = 90°) + d\sigma(\phi = 0°)} \tag{3}$$

solving it gives,

$$Q = \frac{sin^2\theta}{\frac{E_0}{E'} + \frac{E'}{E_0} - sin^2(\theta)} \tag{4}$$

To detect the azimuthal distribution of the scattered Compton photon it is required to have a 2D, or even better, 3D segmented detector. Then by azimuthally binning the position of interaction via photoelectric effect of the scattered Compton photon on the XY plane one can fit the following expression:

$$N(\phi) = Acos[2(\phi - B)] + C \tag{5}$$

where A is the amplitude of the modulation curve, B the polarization angle and C the height of the curve. If the bin counts are normalized to the mean counts per bin, C = 1 and the quality factor Q is defined by,

$$Q = \frac{(C + A) - (C - A)}{(C + A) + (C - A)} = A \tag{6}$$

This technique is very sensitive to methods of determining the XY position of the scattered photon on the detector. If one is using a square pixelated detector several problems arise. The fact that, for example $\theta = 90°$ scattered photons will have longer traveled distance inside the detector if they are scattered diagonally on the azimuthal plane. This fact will increase the probability of photon being reabsorbed on the diagonal azimuthal direction comparing it to photons scattered along the X or Y axis of the square detector, see Figure 2.

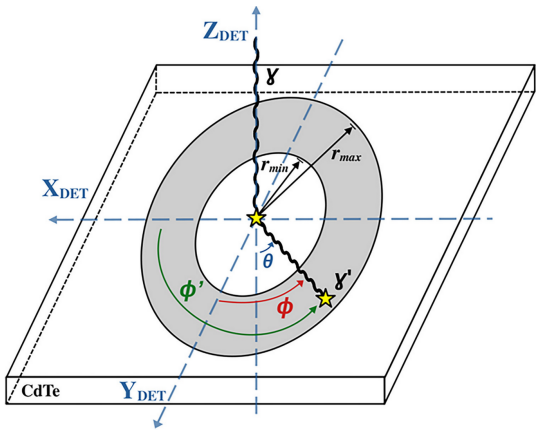


Figure 2: Schematic of Compton scattered photon on a pixelated square semiconductor detector, [Sousa et al., 2026].

This geometrical non-uniformity will induce more hits on the radial bins that are on the diagonal of the detector geometry. This issue is also present at the pixel level. One way of solving the problem is having the detector rotating around its Z axis to uniformize the geometrical induced modulation. This sometimes is not practical. Another way is to perform a renormalization of the polarized induced modulation curve with the detector response to a non-polarized photon beam for the same acquisition conditions. Since this effect will be present in both the polarized and non-polarized modulation we can compute the true counts per azimuthal bin with,

$$N_{true}(\phi) = \frac{N_{pol}(\phi)}{N_{non-pol}(\phi)} N_{non-pol-max} \tag{7}$$

Below it is presented examples of a detector configuration of a modulation curve response to a 100% polarized, non-polarized source and the corrected modulation curve.

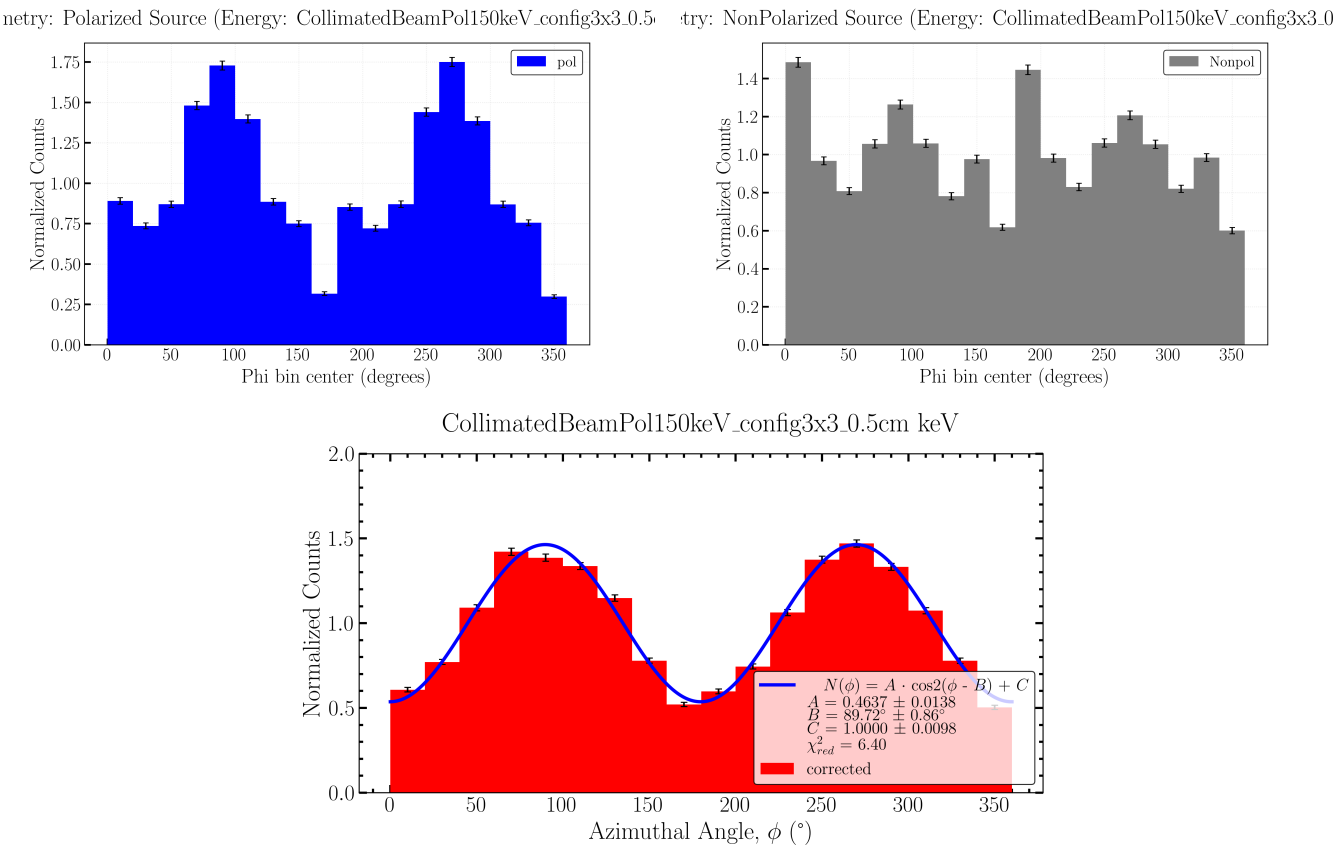


Figure 3: Simulation of 150 keV collimated source hitting PHEMTO Phase 2 configuration. Top left: 100% polarized photons. Top right: Random polarized photons. Bottom: Result from the binning normalization from equation 7

Another issue is when using the discrete pixel location for the determination of photon interaction coordinates. The discretization of location of interaction will limit the radial binning size. This effect is more dominant for Compton events that are separated by a small distance, r_{min} on Figure 2, typically 1-2 pixel distance. We can understand that for Compton events that have adjacent activated pixels have a limit of ϕ azimuthal angles when considering the absolute geometrical location of each pixel on our geometry. What is typically done is either implementing sub-pixel localization techniques on the readout of the detector or one should ignore Compton events that are separated by less than 1-2 pixels.

To determine the fraction of photons that have specific polarization angle on a population of incoming photons we first need to understand what is the modulation response of the detector to 100% polarized gamma-ray beam. This can either be done using simulations or real measurements, preferably via experimental calibration). Having the Q_{100} detector response we can compare the polarization modulation of a given photon population, Q, and comparing it to the Q_{100} thus giving us the fraction of polarized photons,

$$PF = \frac{Q}{Q_{100}} \tag{8}$$

1.2 Laue Lenses

Laue lenses exploit the interference between the periodic nature of light and a periodic structure such as the matter in a crystal. The photons can diffract on a crystal structure and constructively interfere at a specific scattered angle as a function of the incoming light wavelength and crystal structure lattice size. Typically people use Si or Ge crystals due to their structure uniformity.

The orientation of planes in a lattice can be represented through the Miller indices (hkl), whose reciprocals give the fractional intercepts which the chosen lattice plane makes with the crystallographic axes a, b, and c. The spacing between adjacent lattice planes (hkl) is given by d_{hkl} , while the volume of the unit cell is given by V. For a cubic cell with side length a [Frontera, 2025]:

$$d_{hkl} = \frac{a}{h^2 + k^2 + l^2}, V = a^3 \tag{9}$$

For Si a = 5.431 A.

1.2.1 Bragg Law

The photons follow the Bragg-law when they are diffracted. The spacing between the crystal planes d_{hkl} relates to the photon incident angle θ_B and its energy E:

$$2d_{hlk} sin\theta_B = n\lambda = n\frac{hc}{E} \tag{10}$$

where n is the diffraction order, typically they use the first order since its the most efficient [Frontera, 2025], $hc = 12.4\,\mathrm{keV\AA}$, with d_{hkl} and λ in Å and E in keV units [Frontera, 2025].

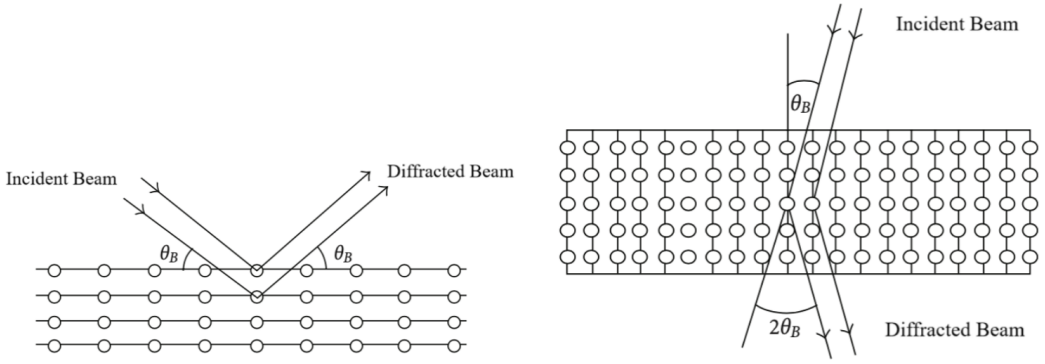


Figure 4: The Bragg condition for constructive interference of a gamma-ray photon with the atoms of a given crystalline plane. Left: Diffraction in reflection configuration also known as Bragg geometry. Right: Diffraction in transmission configuration, also known as Laue geometry [Frontera, 2025]

1.2.2 Geometry of Laue lens

A Laue lens is made of a large number of crystal tiles with diffracting planes in transmission configuration disposed on a spherical cup of radius R, with their diffracting planes perpendicular to the sphere, see Figure 5.

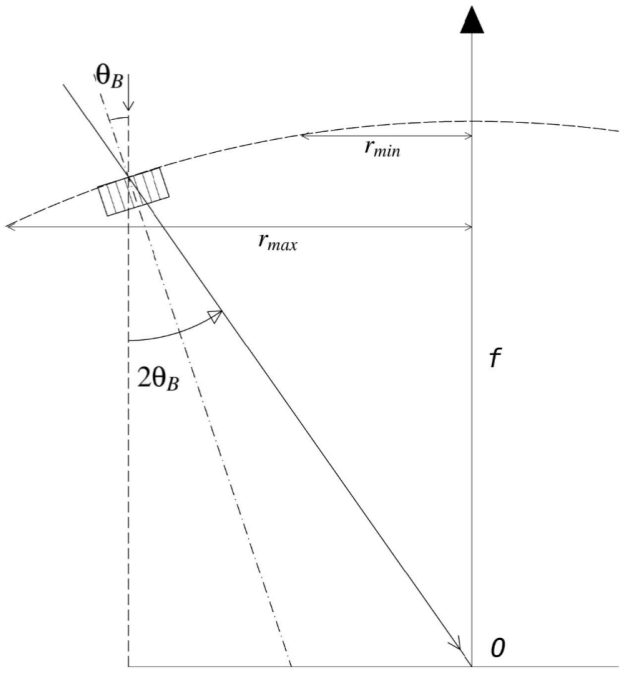


Figure 5: Laue lens geometry. The diffracting crystals on the cup are oriented in such a way to focus in the point O (lens focus) [Frontera, 2025].

The focal spot is at a distance $f = R/2$ from the lenses cup. Equation 10 requires that each crystal has to be oriented in such a way that the angle between the incident beam and the crystalline planes is given by the Bragg angle. Therefore, in order to focus a photon with energy E, for the first order diffraction which is the most efficient, the crystal has to be at a distance r from the symmetry axis given by:

$$r = f \sin(2\theta_B) \approx f \frac{hc}{d_{\mathrm{hkl}} E} \tag{11}$$

This formula is valid for small diffraction angles, with $\cos(\theta_B) \approx 1$. Then it can be derived the formula that gives the diffracted energy E for a given r distance from the center of the lenses:

$$E \approx f \frac{hc}{d_{\mathrm{hkl}} r} \tag{12}$$

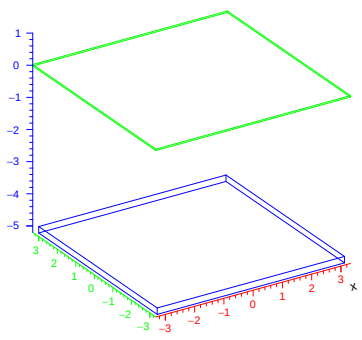
1.3 MEGALIB: HED Thickness - V1

Study the impact on the sensitivity by using a thicker CZT detector with the following configurations for energies 50 keV - 600 keV:

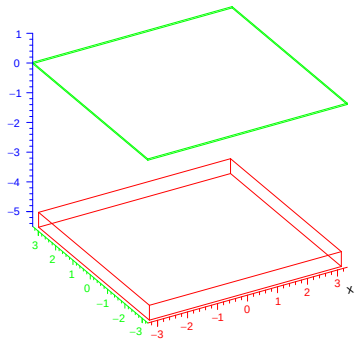
Config 1 Old config, Si top, CdTe bottom 2 mm

Config 2 Si top, CZT bottom 5 mm

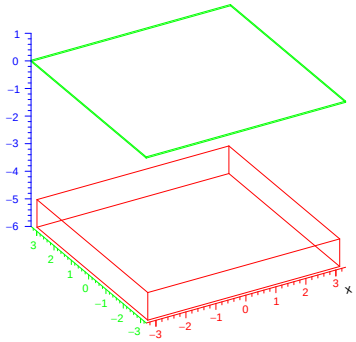
Config 3 Si top, CZT bottom 10 mm



(a) PHEMTO config1. Green is the Si detector 450 µm thick, Blue is the CdTe detector 2 mm thick. The detectors are spaced by 5 cm.



(b) PHEMTO config2. Green is the Si detector 450 µm thick, Blue is the CdTe detector 5 mm thick. The detectors are spaced by 5 cm.



(c) PHEMTO config3. Green is the Si detector 450 µm thick, Blue is the CdTe detector 10 mm thick. The detectors are spaced by 5 cm.

Figure 6: PHEMTO preliminary configurations for MEGAlib simulations.

For the current simulations the following detector characteristics were used:

Detector	Parameter	Value	Unit
Si	Area	6.65×6.65	cm^2
	$\sigma_x\sigma_y$ resolution	0.014	cm
	ρ_{Si}	2.33	$\frac{g}{cm^3}$
	$E_{threshold}$	2	keV
	$E_{resolution}$	$\sigma = 0.15$ @ 4	keV
		$\sigma = 0.15$ @ 6	keV
CdTe	Area	6.4×6.4	cm^2
	$\sigma_x\sigma_y$ resolution	0.025	cm
	ρ_{CdTe}[1]	5.78	$\frac{g}{cm^3}$
	$E_{threshold}$	4	keV
	$E_{resolution}$	10.6% @ 40	keV
		4.7% @ 100	keV
		1.2% @ 500	keV
$C_{0.9}Z_{0.1}T$	Area	6.4×6.4	cm^2
	$\Delta x\Delta y$ resolution	0.025	cm
	ρ_{CZT}	5.78	$\frac{g}{cm^3}$
	$E_{threshold}$	4	keV
	$E_{resolution}$	10.6% @ 40	keV
		4.7% @ 100	keV
		1.2% @ 500	keV

Table 1: PHEMTO preliminary simulation detectors assumptions.

For this study we only considered the photons coming from the Laue Lenses optics. We studied the instrument detection efficiency for the energy range 50-700 keV. A collimated 1.5 mm diameter monochromatic beam was generated for each energy. A total of 10^6 events were generated for each energy. The beam was aimed to the geometrical center of the detectors, see Figure 7.

[1] Used the CZT density

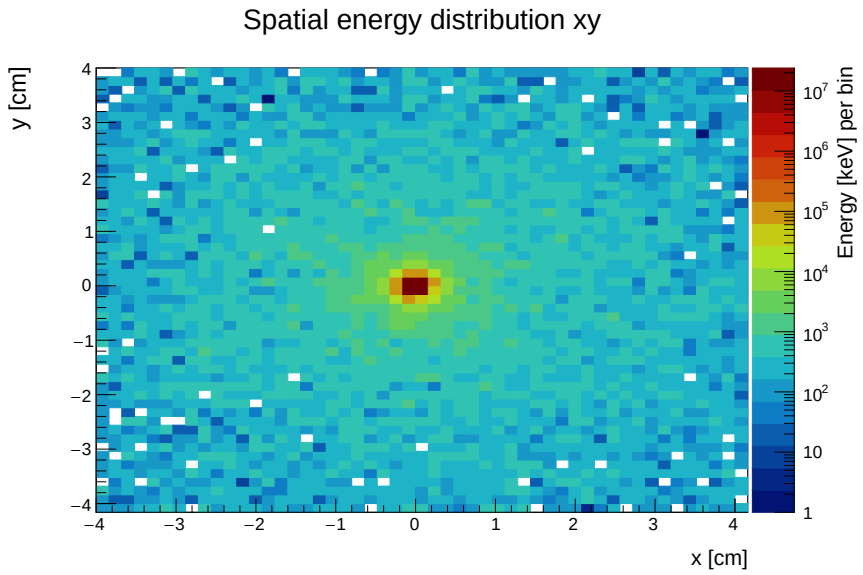
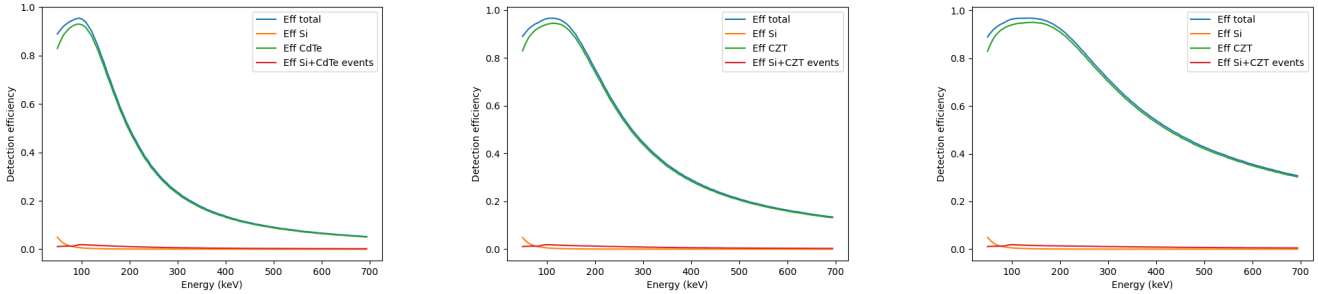


Figure 7: Example of the 2D energy distribution on the config1 detectors for a 104 keV collimated, 1.5 mm diameter, source.

Figure 8 showcases the simulated detection efficiencies for the different instrument configurations.



(a) PHEMTO config1, CdTe 2 mm, simulated efficiency for the energy range 50-700 keV.

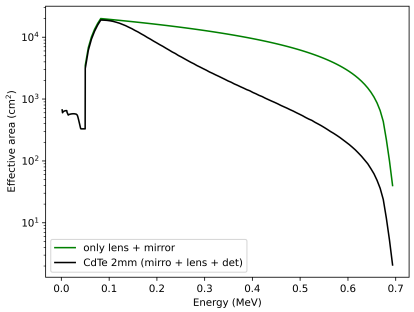
(b) PHEMTO config2, CZT 5 mm, simulated efficiency for the energy range 50-700 keV

(c) PHEMTO config3, CZT 10 mm, simulated efficiency for the energy range 50-700 keV

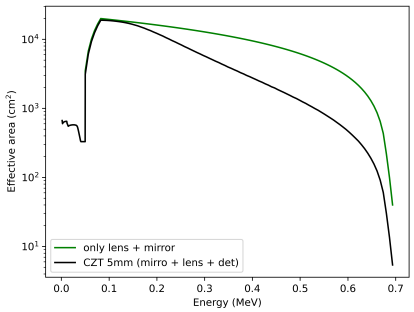
Figure 8: PHEMTO instrument detection efficiency for different configurations

For energies below 50 keV I used the data from Dr.Phillipe Laurent analysis.

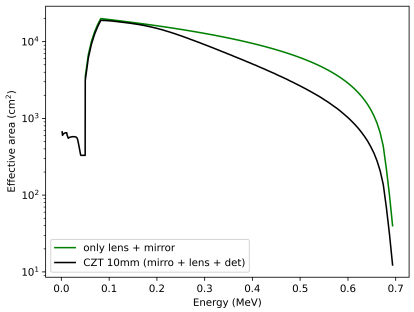
Regarding the Laue Lenses efficiencies, I used the effective area provided by Dr.Phillipe Laurent. The Figure 9 showcases the effective area of the lenses (laue + x-ray mirror) as well as the effective area of the whole system, with the detector.



(a) PHEMTO config1, CdTe 2 mm, instrument effective area.



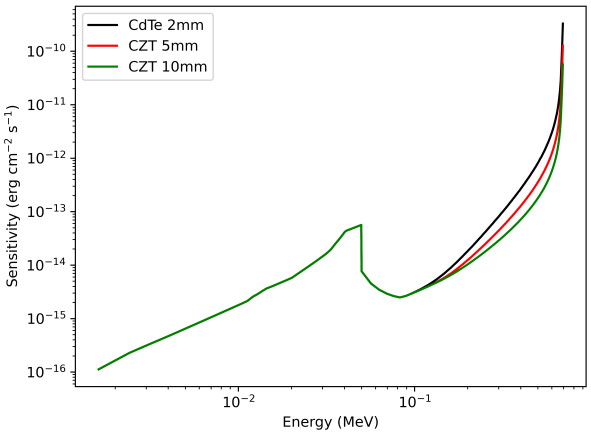
(b) PHEMTO config2, CZT 5 mm, instrument effective area.



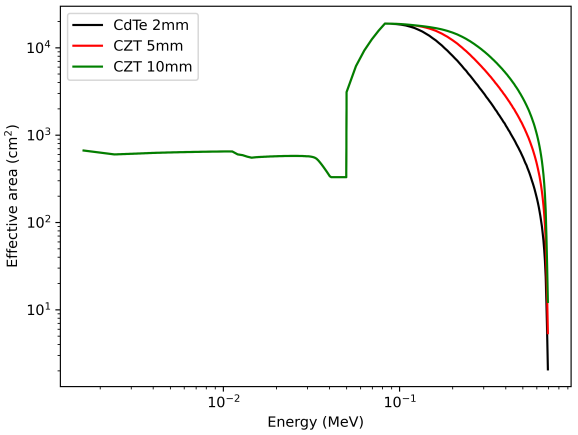
(c) PHEMTO config3, CZT 10 mm, instrument effective area.

Figure 9: PHEMTO instrument detection efficiency for different configurations

With this, I used the excel tool developed by Dr.Phillipe Laurent to compute the instrument sensitivity in $ergcm^{-2}s^{-1}$. The Figure 10a showcases the sensitivity comparison for the different PHEMTO configurations. Figure 10b showcases the effective area comparison for the different PHEMTO configurations.



(a) Comparison of the PHEMTO configurations sensitivity.



(b) Comparison of the PHEMTO configurations effective area.

Figure 10: PHEMTO instrument detection efficiency for different configurations

In order to understand how PHEMTO sensitivity compares with the most sensitive detectors in orbit we adapted a figure from [Lucchetta et al., 2022] to include the PHEMTO config3, 10 mm CZT, see Figure 11.

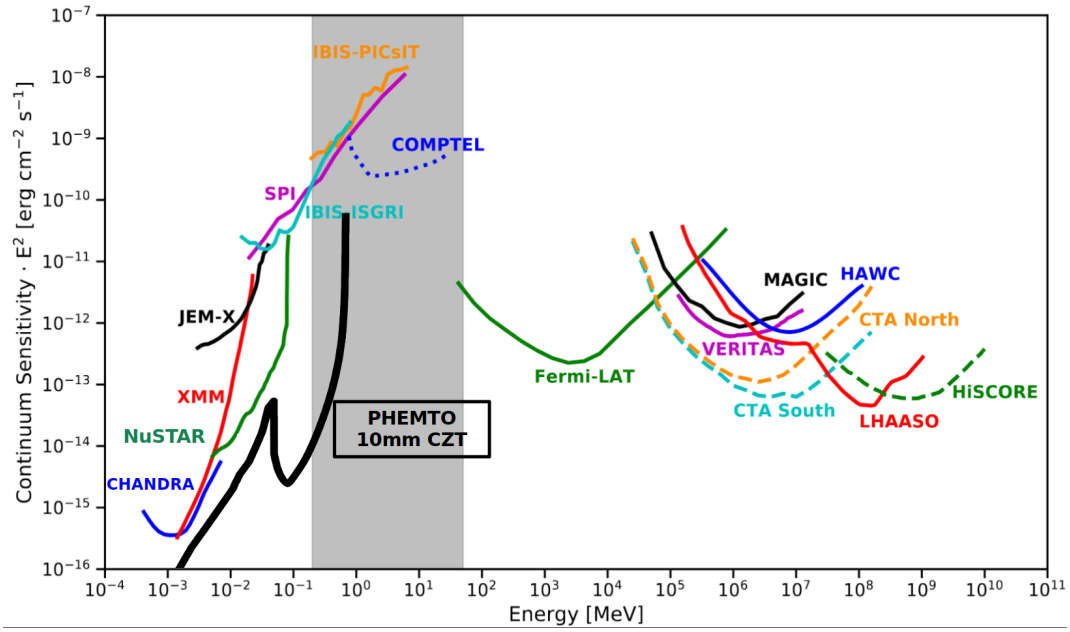


Figure 11: PHEMTO config3, 10 mm CZT, sensitivity comparison with other instruments current in orbit, except INTEGRAL instruments (SPI, IBIS). Adapted from [Lucchetta et al., 2022].

1.4 MEGALIB: Polarimetry Vs. HED size and LED-HED distance - V2

Considering the LED and HED characteristics presented on table 1 a study on the impact on polarimetry performance. For a fixed LED detector configuration we varied the size of the HED from 4x4 CdTe 2 mm detector matrix up to 7x7 matrix. Regarding the vertical distance we varied the distance between the bottom side of the LED to the top side of the HED from 0.5 cm up to 9.5 cm with a step of 1 cm.

To estimate the instrument capability of detecting a % of polarized photons within a population with a confidence level of 99% the Minimum Detectable Polarization (MDP) metric is often used [Weisskopf et al., 2010]. The MDP is given by:

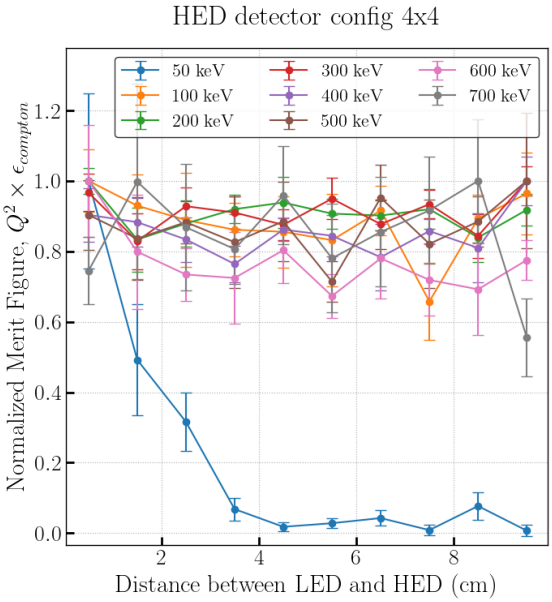
$$MDP = \frac{4.29}{\mu_{100}\epsilon_{Compton}N_S}\sqrt{\frac{\epsilon_{Compton}(N_S + N_B)}{T}} \tag{13}$$

where μ_{100} is the modulation amplitude for a 100% polarized photon source, $\epsilon_{Compton}$ is the Compton event identification efficiency, N_S is the source photon flux irradiation the detector system and N_B is the background rate at the PSF area. PHEMTO optical system configuration allows $N_S \gg N_B$, therefore we have the following figure of merit F:

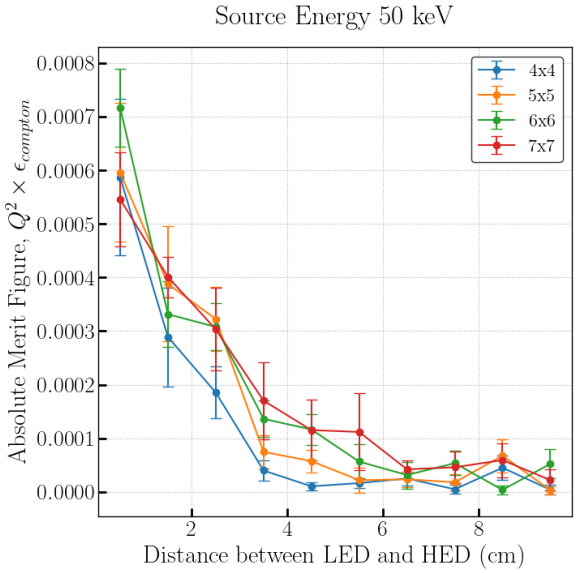
$$F \propto MDP^2 \propto \frac{1}{\mu_{100}^2\epsilon_{Compton}} \tag{14}$$

We will use this figure of merit to compare the different configurations of PHEMTO given that the N_S and N_B are the same for all the configurations.

From the analysed data the first major conclusion is that the polarimetric performance of the PHEMTO configuration is better as the LED and HED are closer to one another, Figure 12b. This impact is more evident for lower energies where the Compton cross section of the Si is higher. As the photon energy increases the 450 µm Si becomes transparent to the photons and therefore we only rely on the HED to perform polarization measurements.



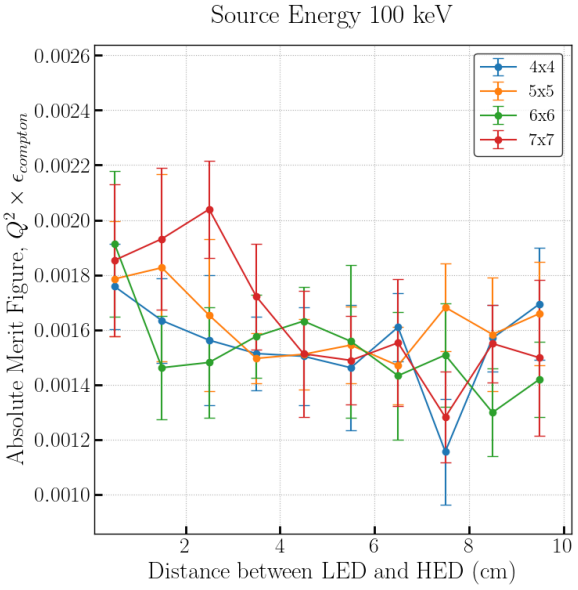
(a) Normalized inverse of Merit Figure, $\frac{1}{F}$, for 4x4 HED configuration as a function of the distance between LED and HED. Notice that this plot only showcases how F behaves when varying the distance between detectors. The plot does not showcase the instrument polarimetric performance between different energies.



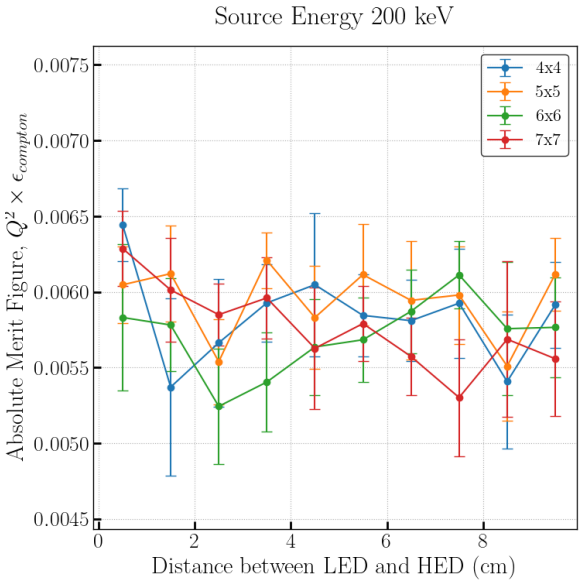
(b) Inverse of the Merit Figure, $\frac{1}{F}$, using absolute compton efficiency, as a function of distance between LED and HED for a 50 keV collimated source. Figure showcases the negative impact on polarimetric performance when distance between detectors increases. Also, the size of HED doesn't seem to have an impact on polarimetric performances for distance <1.5 cm, **further study needed**.

Figure 12: Polarimetric performace of PHEMTO detector configuration. Notice that it is presented the inverse of the Merit Figure, F, from equation 14, therefore, the higher the y-axis the better instrument polarimetric performance.

Figure 12a also showcases that for higher energies, $E_\gamma > 100$ keV, there is no clear impact on the polarimetric performance of the instrument as the distance between detectors increase. This can more easily be seen on Figures for energies of 100 keV and 200 keV.



(a)



(b)

Figure 13

Taking these results into account, for the current design it is recommended that the distance between detectors should be lower than 1.5 cm and that the size of the HED detector should be defined taking into account imaging reconstruction capabilities given by the optics design.

1.5 MEGALIB: ESA PHASE 2 simulation - V3

1.5.1 Detectors Assumptions

At this stage of the PHEMTO design the detectors performance is not yet defined. Although PHEMTO aims to operate between the energy range of 1-400 keV, during the simulation process we extended the operational energy range up to 600 keV.

LED - Low Energy Detector

The LED is a monolithic silicon pixel detector. It will consist of an array of 256 × 256 DEPFET (DEpleted P-channel MOSFET) macropixels with a pixel size of 250 × 250 μm^2 each, integrated on a high resistivity silicon bulk with 450 μm thickness. The final sensitive area of the detector will be 64 × 64 mm^2.

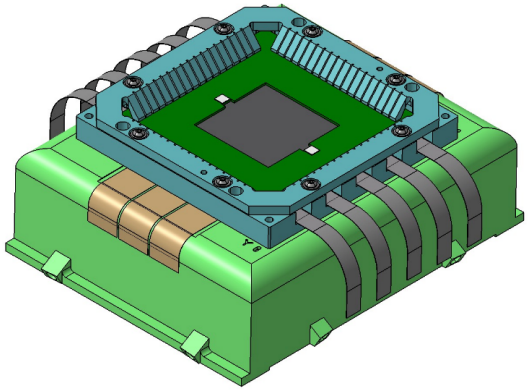


Figure 14: PHEMTO/LED CAD model.

The LED is read out in 1-fold column parallel rolling shutter mode, using 4 specimen of the analog amplifier / filter ASIC VERITAS II. A row readout time of 2.5 μs is targeted, which yields a total exposure time for each frame of 640 μs. **At this stage of the simulation chain, these readout timings were not considered and therefore we assumed zero dead time**.

For the LED, we assumed an end-of-life energy resolution of 150 eV across the full energy range as stated from the energy resolution degradation study performed by the LED team, see Figure 40.

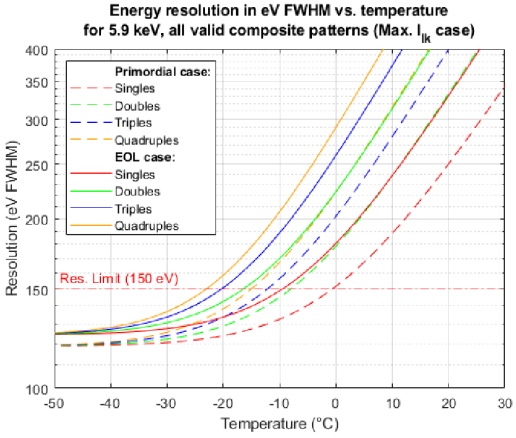


Figure 15: PHEMTO/LED energy resolution at the beginning and end of life vs the operation temperature. For the end of life the LED team used the hypotheses of 10^6 10 MeV-equivalent proton damage, 250 pA/cm^2 leakage current and 5e^- rms ENC.

HED - High Energy Detector

The HED layer will be based on Cd(Zn)Te pixel detectors. The detection layer is a 3x3 mosaic of individual CdTe crystals of 24.5 x 24.5 mm^2 of active area. Each module has the size of 26 mm and the inter module gap is 0.5 mm. Each active area of the crystal is segmented by a 96 x 96 pixel readout with 250 μm square pixel size. The total detection area is 79 x 79 mm^2, creating a matrix of 288 x 288 pixels. **At this stage of the simulation I used a monolithic crystal of** 7.2 mm^2 **with 288 x 288 pixels,** 250 μm.

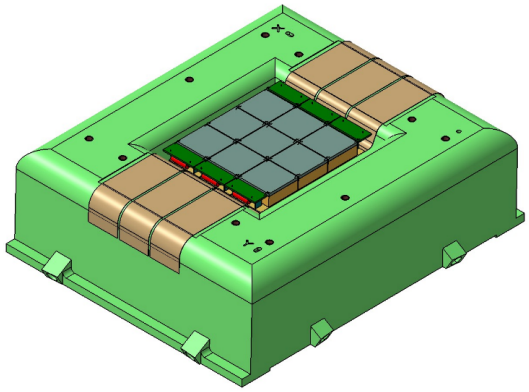


Figure 16: PHEMTO/HED CAD model.

For the HED we assumed the noise level of the current status of the technology, 74e^- up to 200 keV and a constant 0.5% in the range 200-600 keV, see Figure 17.

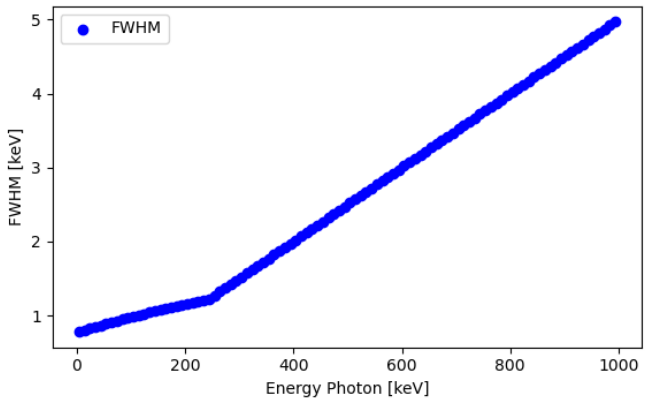


Figure 17: PHEMTO/HED energy resolution as a function of energy. 74 e^- until 200 keV and 0.5% from E > 200 keV.

1.5.2 Optics Assumptions

Laue lenses focusing technology is still in development and the current best experimental results do not achieve the angular resolution requirements of $\approx 25''$ at 200 keV. [Ferro et al., 2024] experimentally, with some extrapolation with the use of simulations, showcased a best case scenario of angular resolution of 289$\pm$ 4 arcsec. At this stage the authors are investigating assembly techniques to reduce the Bragg misalignment and in turn improve the angular resolution. Currently, for PHEMTO we can assume the study case presented on [https://larixfacility.unife.it/wp-content/uploads/2019/02/Valsan_PhD_Thesis.pdf]. The authors used Ge(111) and at this stage we extrapolate that PHEMTO PSF will follow the authors values, although for PHEMTO we are also using Si(110) for the low energy part. From the authors results we can approximately assume that the PSF half-power diameter is 1.5 mm. From [Khalil et al., 2015], where they had similar assumption, they deduced that in order to reconstruct the PSF shape the detector plane 2D spatial resolution should be better than 500 μm.

For the current study we generated a collimated photon beam hitting the detector azimuthal plane with polar angle $\theta = 0°$ with a diameter of 1.5 mm. At this stage we are also assuming that the design of the X-ray mirror

will try to mimic the PSF of the Laue Lenses and therefore the PSF on the mirror energy band is the same as the PSF for the Laue Lenses.

Regarding the optics effective area, the values from older ASTENA observatory proposal were used. The effective area is presented in the figure 18.

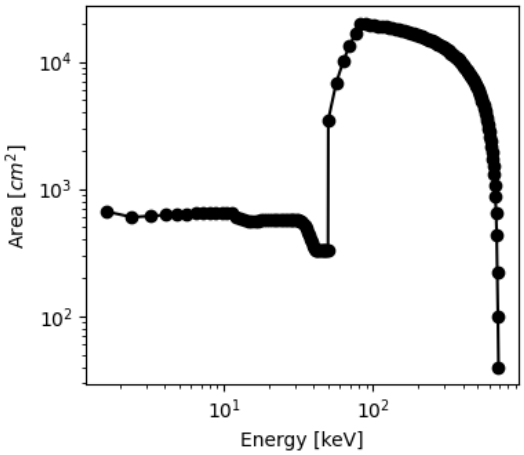


Figure 18: PHEMTO Phase 2 effective area. For both x-ray mirror and Laue Lenses system. Values taken from previous ASTENA observatory proposal. Values to be confirmed by the respective teams for future analysis.

1.5.3 Low Energy Polarimetry Vrs LED-HED distance

As it was demonstrated on Figure 12a polarimetry at low energy, <100 keV is highly impacted by the distance between the LED and HED. Now with the defined LED and HED configuration (thickness, pixel size, mosaic size) it is important to define the vertical separation distance according to both polarimetry performance and mechanical constraints.

To study the polarimetric performance of the system the approach presented on Section 1.4. Now we used monochromatic beam with energies from 5 to 100 keV. We used MEGAlib toolkit to simulate the different focal plane geometries from inter-plane detector distance of 0.5 cm up to 5 cm. It was noticed that the low energy limit for polarimetry is 15 keV since for lower energies the Compton cross section of the Si is too low and therefore the Compton events detection efficiency is ≈ 0.

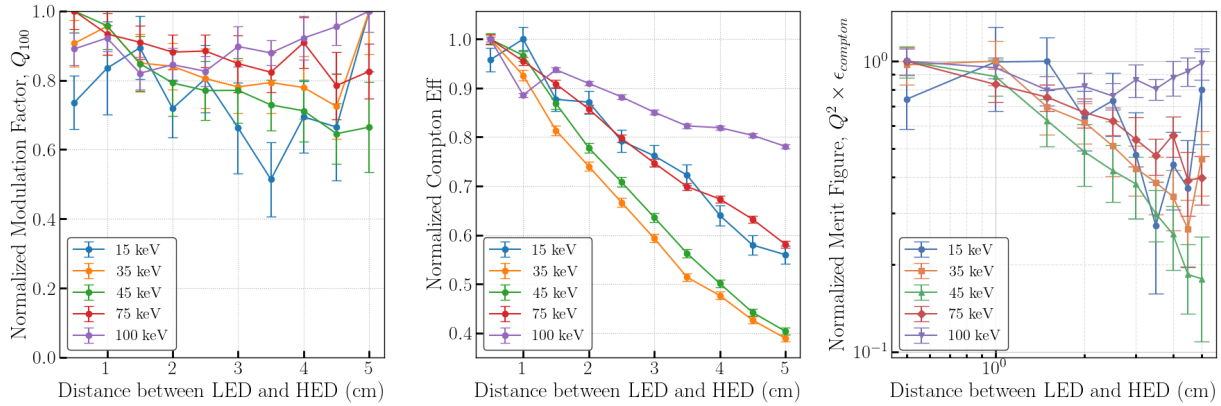
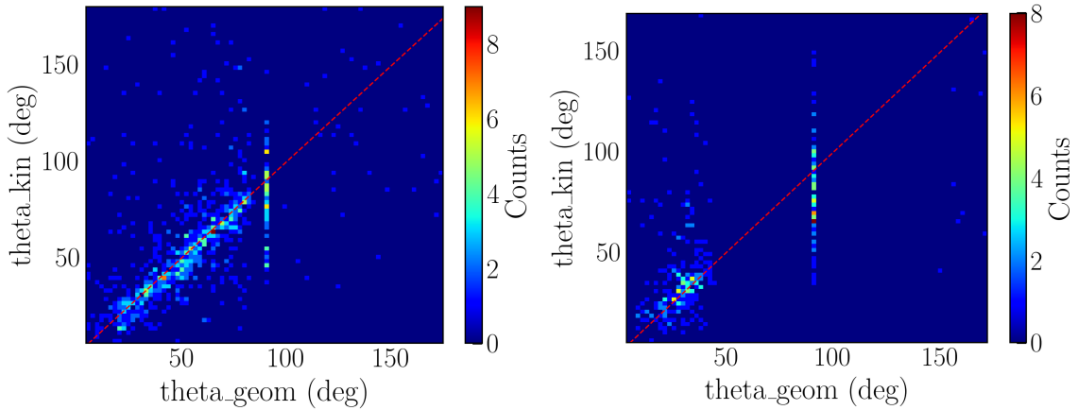


Figure 19: PHEMTO focal plane polarization performance as a function of the distance between LED and HED.

From Figure 19 it is clear that for polarimetry at low energies it is advantageous to have the detectors as close to one another as possible. This can be understood by the increase in Compton events detection efficiency. From Figure 19 left we can see that for 100 keV as the distance interplane increases the Q_{100} modulation factor also increases. This is expected since we saw from equation 4 the modulation factor depends of the θ scattering angle of the Compton event. For 100 keV the modulation factor is a maximum for $\theta \approx 90°$ and it is lower for any other

θ scattering values. Therefore in the case when HED and LED are very close to each other we are accepting many Compton events with $\theta \neq 90°$ therefore reducing the modulation factor. As we then increase the distance between the detectors, the ratio of Compton events at $\theta \approx 90°$ is higher since some of the scattered Compton events now miss the HED detector (solid angle of coverage is lower as we increase the HED LED distance-need to explain this better). We also may attribute this reasoning to the sudden jump on the Q_{100} modulation factor for 15 keV when the inter detector distance is at 5 cm - in this case we are basically only performing polarimetry on the Si detector with Compton events with θ very close to 90°. We can see this effect on the example for 55 keV source on Figure 20.



(a) 55 keV, LED to HED distance of 0.5 cm. (b) 55 keV, LED to HED distance of 5 cm.

Figure 20: 2D Histrogram of the geometrical determined θ scattered angle using the detectors spacial resolution and the θ scattering angle determined by the Compton kinematics equation.

Taking these results into account, for the current design it is recommended an inter-plane distance of 1 cm. **For future analysis one should put this in terms of the impact on observation time to achieve a determined MDP for the Crab or other sources. This way, and by studying the inter-plane distance with millimeter resolution one could give the detectors team real impact on the performance of the observatory and maybe push for stricter or more relaxed mechanical requirements.**

1.5.4 Minimum Detectable Polarization (MDP)

To access the polarimetric performance of the PHEMTO instrument configuration a Monte Carlo simulation study was performed using the MEGAlib toolkit, [Zoglauer et al., 2006], for the energy range 15 keV-400 keV since for $E_\gamma < 15$ keV the Compton cross section of the LED Si material is very low. The inter detector distance was set to 1 cm.

To estimate the instrument capability of detecting a % of polarized photons within a population with a confidence level of 99% the Minimum Detectable Polarization (MDP) metric is often used [Weisskopf et al., 2010]. The MDP is given by equation 13 where μ_{100} is the modulation amplitude for a 100% polarized photon source, $\epsilon_{Compton}$ is the Compton event identification efficiency, N_S is the source photon flux irradiation the detector system and N_B is the background rate at the PSF area.

In order to correct for systematic, geometry-induced modulations in the azimuthal distribution of Compton events, two photon populations were simulated: fully (100%) polarized photons and background non-polarized photons.

At this stage we have two approaches when computing a source MDP. The first approach is simulating the Q_{100} and $\epsilon_{Compton}$ for specific monochromatic energies using monochromatic sources in MEGAlib. Then for MDP computation we integrate the background and source fluxes with the center bin on the simulated Q_{100} and $\epsilon_{Compton}$. This method aproximates the polarimetic performance of the detector over a binned energy range, so, the finner the energy bin the better the approximation is to the polarimetric performance of the detector. The second method is closer to reallity. Having a differential source flux model or a table with the differential or binned flux we can creat a flux model to input directly on the MEGAlib simulation. It is important to note that the flux that we need to simulate on MEGAlib is the flux hitting the detector therefore we need to fold the source flux with the optics effective area,

$$\Phi_{bin} = \int_{E_{min}}^{E_{max}} A(E)\Phi(E)dE \tag{15}$$

where Φ_{bin} is the source flux for a given energy bin $[E_{min}, E_{max}]$, the $A(E)$ is the optics effective area and $\Phi(E)$ is the source differential flux in units $[ph/s/cm^2/keV]$. The way MEGAlib works is that it needs a .dat file that has the source flux morphology normalized to energy, ie $[ph/s/keV]$, and then present the normalization factor that is the total value of ph/s over the interested energy range,

$$\Phi_{total} = \sum_i \Phi_{bin,i} = \sum_i \int_{E_{min,i}}^{E_{max,i}} A(E)\Phi(E)dE \tag{16}$$

From this we get the real flux at the detector level after the optics system. Then we simulate two photon components, 100% linear polarized photons and random polarized photons. From the result of the Monte Carlo simulation we use the Revan tool to perform Compton event identification by computing the Compton θ_{ene} scattering angle using the energies deposited on the detectors and comparing it with the geometrical θ_{geo}. The ^{2} metric is used to accept the Compton event.

$$\chi^2 = \sum_i \frac{(\theta_{ene} - \theta_{geo})^2}{\sigma_i^2} \tag{17}$$

where $\sigma^2 = \sigma_{geo}^2 + \sigma_{ene}^2$, σ from energy and geometry are computed from the error propagation of the Compton kinematics equation accounting for the detectors energy and position resolution[2].

Notice that at this stage we already have the $\epsilon_{Compton} N_s$ product presented on the equation 13 and we can now bin the energy to compute the respective Q_{100} but due to implementation issues[3] we are, at this stage, for each energy bin we are computing the relative Compton efficiency and the corresponding Q_{100}. Then we re-computing the source flux at the detector level for a the given energy bin to get the absolute Compton event detection efficiency.

$$\epsilon_{Abs_{Compton-bin}} = \frac{N_{Compton-MEGAlib}}{N_{sim-events} \frac{\int_{bin_{min}}^{bin_{max}} A(E)\Phi(E)dE}{\int_{E_{obsMin}}^{E_{obsMax}} A(E)\Phi(E)dE}} \tag{18}$$

where bin_{min} and bin_{max} are the edges of the Energy bin, E_{obsMin} and E_{obsMax} are the minimum and maximum operation energy range of the observatory, $N_{Compton-MEGAlib}$ is the number of Compton events reconstructed from MEGAlib:Revan for the given energy bin and $N_{sim-events}$ is the total number of simulated photons on MEGAlib:Cosima Monte Carlo simulation. Regarding the background flux, at this stage we are only taking into account cosmic photons for a LEO orbit of 650 km with an inclination of 5°, [Cumani et al., 2019]. For each energy bin we are considering photons coming from 2π sr and we get the background counts by folding the flux to the PSF area and applying the $\epsilon_{Abs_{Compton-bin}}$.

[2] details are left as an exercise for the reader.
[3] Code is not properly organized, lack of time

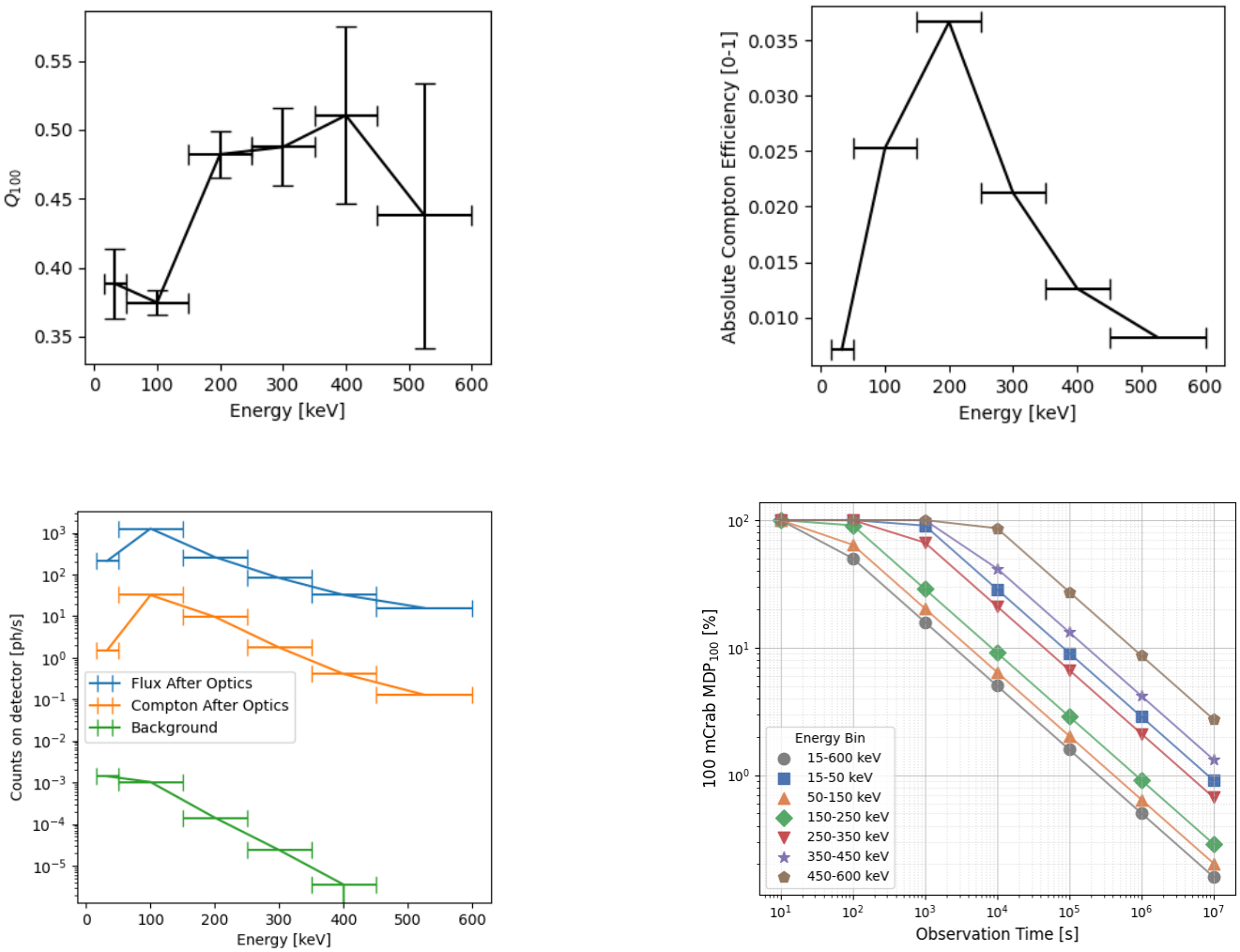


Figure 21: Performance metrics for the PHEMTO configuration observing a 100 mCrab source across different energy bins. Top left: Q_{100} computed for each energy bin accounting for source flux at detector level. Top right: Compton detection efficiency for each energy bin. Bottom left: Source and bakground event count rate at detector level, 1Crab. Bottom right: Computed MDP for 100mCrab source.

The current approach allows the user to give source fluxes either as a model[4] or via differential flux tables. Below we present the MDP for the very bright supernova remnants sources V0332 and 1A0535.

[4]In the future we will implement capability of importing source models via XSpec

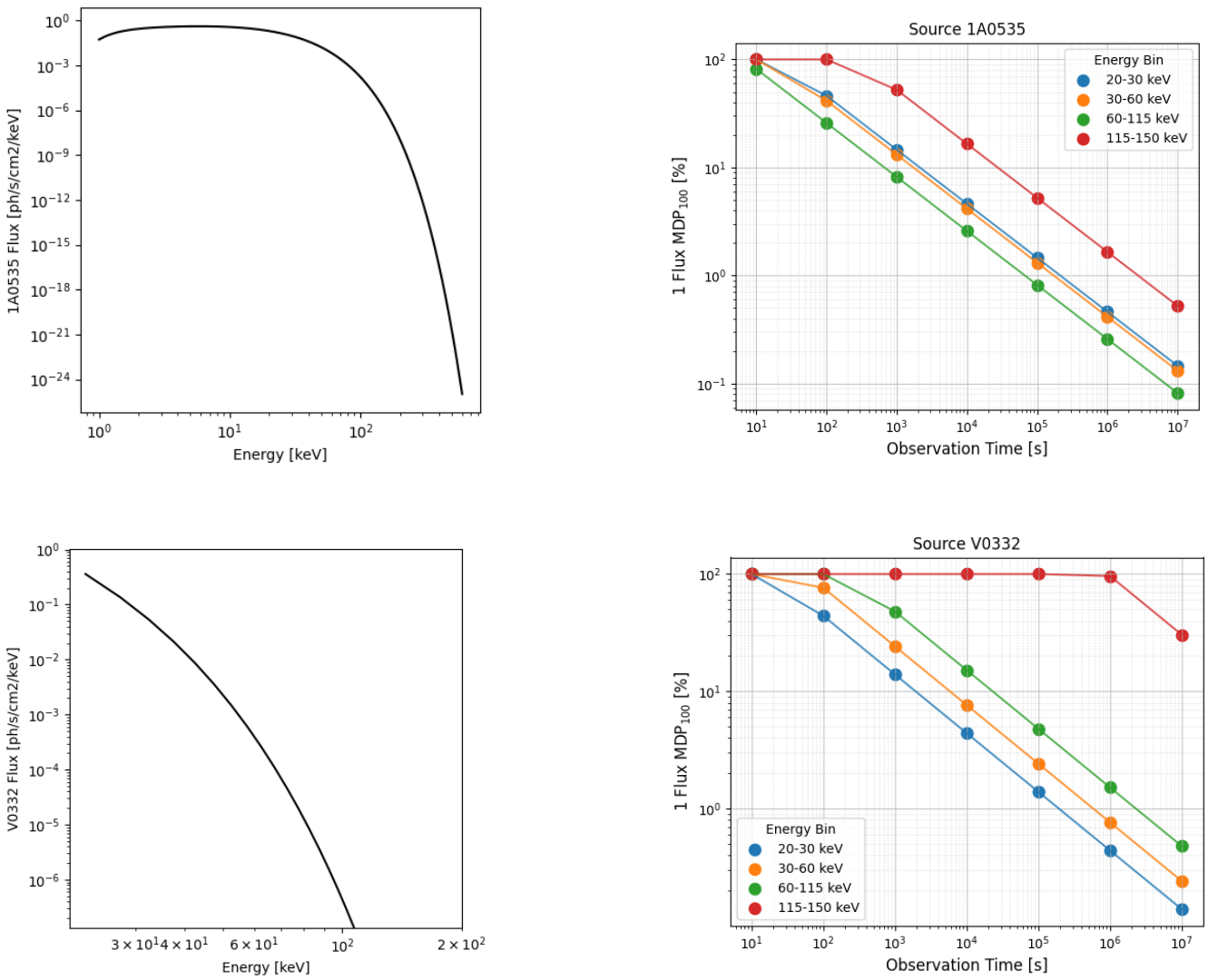


Figure 22: Performance of PHEMTO configuration observing a super novae remenets 1A0535 and V0332 sources. Top left:1A0535 source flux. Top right: MDP for the 1A0535. Bottom left: V0332 source flux. Bottom right: MDP for the V0332.

1.5.5 Sensitivity - V2.2

For the set detector configuration a sensitivity study was done. The sensitivity of the instrument is very dependent on the background model we use. To reduce the cosmic X-ray background component for lower energies on the PSF area of the detector system, we will use a 0.13 cm thick lead (Pb) cylindrical collimator positioned 5 cm above the LED. Taking into consideration the lens size, viewed at the focal length of 20 m we have determined its aperture angle to be $\approx 5°$, implying a collimator length around 100 cm and an inner radius of 7.6 cm. This configuration will allow >99 % X-ray diffuse background suppression at the PSF area for energies until ≈150 keV.

Efforts have been made to perform a MEGAlib Monte-Carlo simulation taking into consideration Cosmic background fluxes from [Cumani et al., 2019] and considering the Collimator structure, presented in figure 23 left. The current approach was to generate photon flux that hit the PSF area and passed the collimator structure to understand the transparency of the collimator and have a somewhat real approximation of the x-ray background flux. The source geometry generated on MEGAlib was the following:

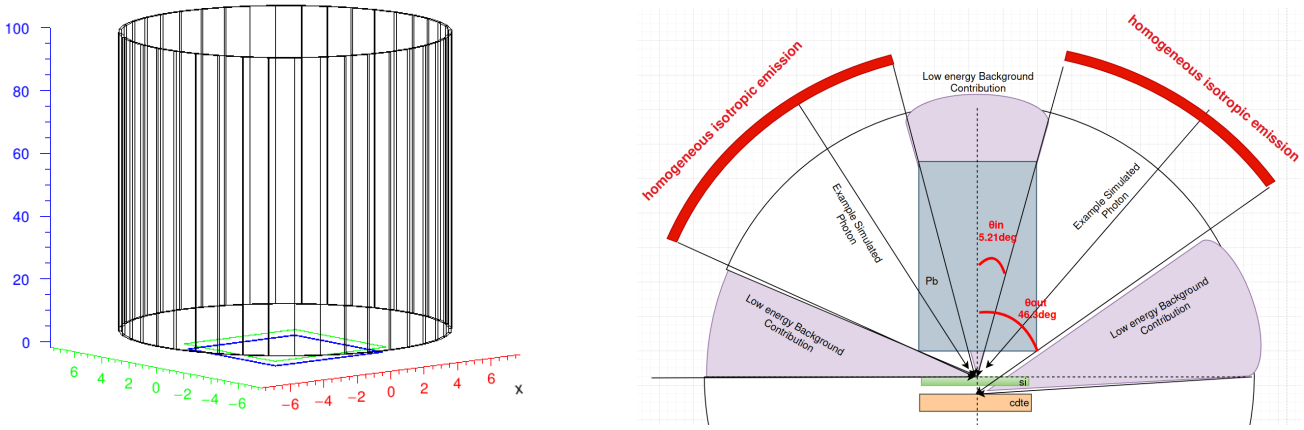


Figure 23: Left: MEGAlib PHEMTO geometry with the lead collimator. Green square is the Si detector, Blue square is the CdTe detector and black structure is the collimator. Right: Background isotropic emission MEGAlib source geometry.

At this stage the simulations have two problems. The first one is that if we want to simulate for long integration times, we need to have enough statistics to reliably compute background fluxes at PSF area. The second is that we are getting too many counts at PSF area for low energies and an unexpectedly high background flux for low energies. This might be a problem with the geometry. There might be a problem when I perform the coordinate transformation from the MEGAlib reference frame to the individual detector reference frame[5]. At this stage we are not using the simulation fluxes for the background computation.

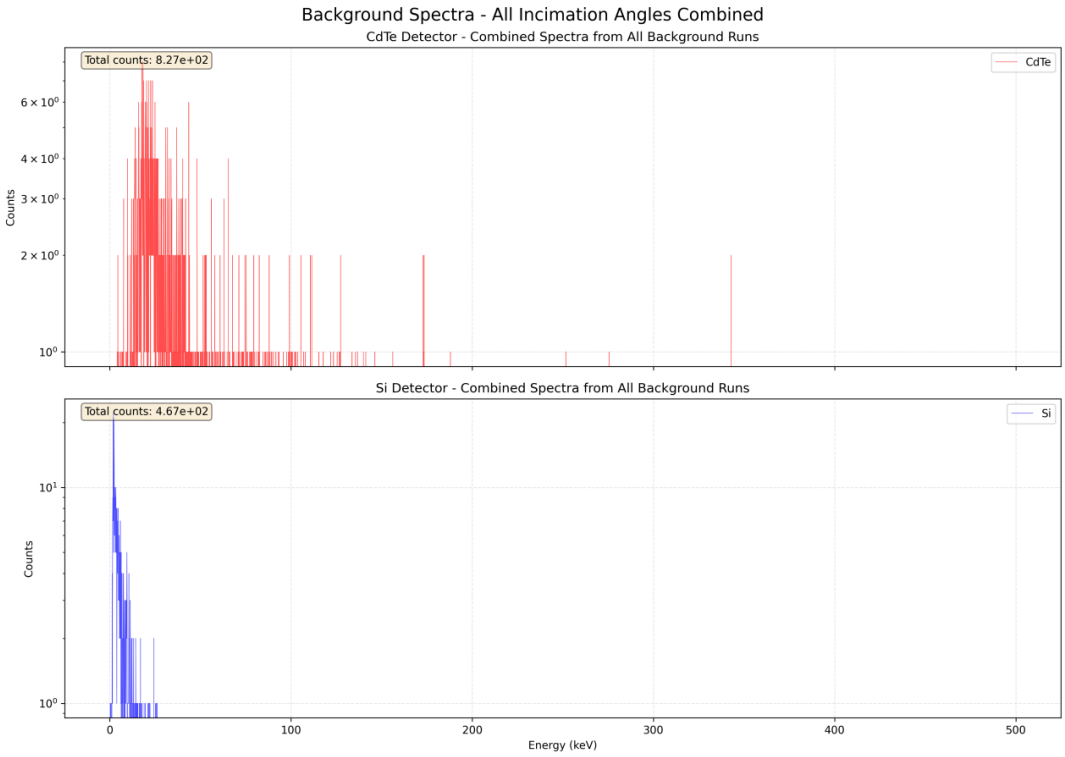


Figure 24

At this stage we are using an approximation made by Dr.Philippe Laurent where he took a cosmic background model and considered the detector FOV, according to collimator geometry, material thickness, and computed the

[5]I need to do this to assign the hits to the respective pixels on the detector

count rate at the PSF area. Afterwards we performed a Monte-Carlo simulation with monochromatic sources to get the detector system photon efficiency over the observatory energy range.

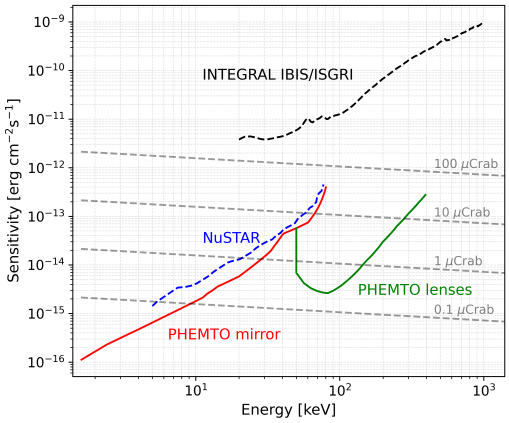


Figure 25: PHEMTO sensitivity curve compared to INTEGRAL IBIS/ISGRI and Nustar obsertatory.

1.6 MEGALIB: RMF and ARF - V4

The objective of this simulation run is to generate XPEC ready RMF and ARF files of the detector system.

1.6.1 MEGAlib physical Geometry

1.6.2 Detector Assumptions

The geometry used in this round of simulations was the same as the one presented in 1.5.

We updated the depth dependent response of the CdTe detector. For the CdTe detector the charge collection efficiency depends on the depth interaction point of the photon. This effect will produce a non-Gaussian shape for monoenergetic incident beam. A tail on the low energy side of the energy peak will develop. As seen in the Paper from NDIP the left side of the energy peak is not only the product of charge loss due to the interaction depth. Also for events with large charge close to the anode the charge can also be induced in side pixel anodes. If the threshold on the anodes is high we will be registering events in a single pixel instead of multiple pixels. Therefore the charge induced on the single pixel will be lower than the energy peak and populate the tail of the energy peak. In order to model these two effects we chose to use an empirical formula instead of the output of CadTool because at this stage we are not fully confident on the full representability of the spectrum shape. We are using the following charge collection efficiency as a function of depth,

$$CCE(z) = \frac{-1}{(50 \cdot (z_{norm} + 1)} + 1.01 \tag{19}$$

where z_{norm} is the normalized distance between $-z_h$, the distance from center of detector to the anode and z_h the distance from center of the detector to the cathode.

In addition MEGAlib as an option to include a Z dependant energy loss. On the .geo one has to include *DETECTOR.EnergyLossMap FILENAME.lossMap.dat.*

Introducing this behaviour we get a more realistic high energy photopeak energy spectrum, see figure bellow.

We will be using this approach to build the instrument RMF. The Si detector suffered no change.

Employing a depth-dependent CCE will in turn introduce an average overall mean charge collected at the anode to be reducing with the energy. This can be understood as the photons for higher energy are more uniformly interacting with the detector depth. Since for photon interactions close to the anode the CCE is very low and as the photon energy goes up we are increasing the probability of this population we are in fact lowering the average CCE per photon energy. This will introduce a non-linearity on the detector energy response that can be linearly calibrated, see Figure 29.

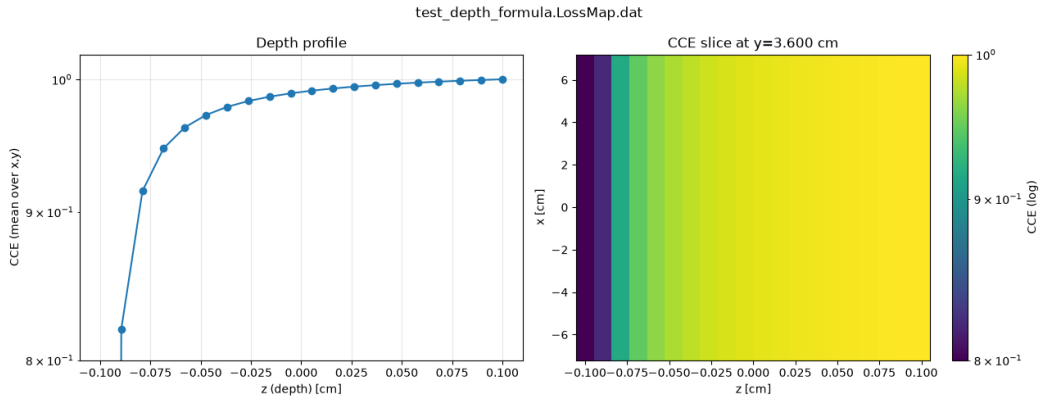


Figure 26: Empirical CEE used on MEGAlib. Made sure that close to cathode the CEE is 1 but in reality this isn't true. This was just to minimize the post-calibration that I would need to do before computing the RMF.

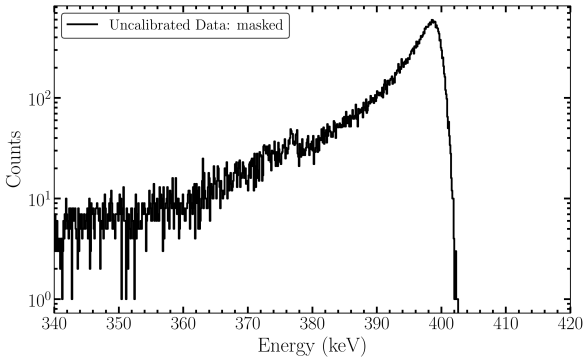


Figure 27: Simulated MEGAlib MC2 spectrum. 0.5% energy resolution with the empirical formula for tailing effect 19.

1.6.3 Optics Assumptions

From updates from Italian colleagues we are following the technology of Si(111) for the Laue Lenses. They showed via simulations that using only Si lenses up to 400 keV revealed similar effective area.

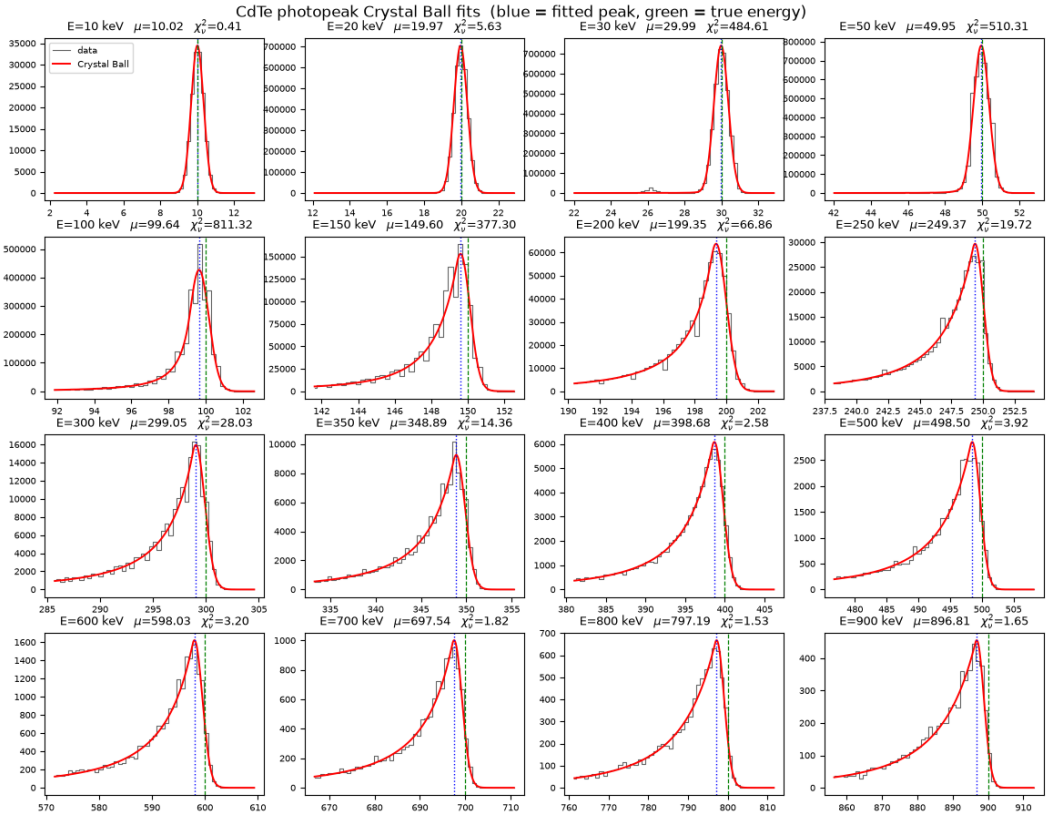


Figure 28

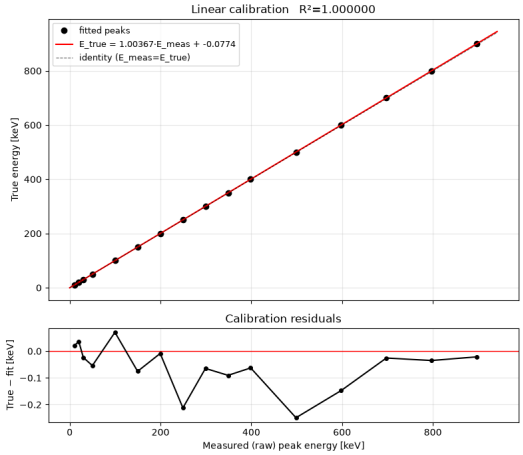
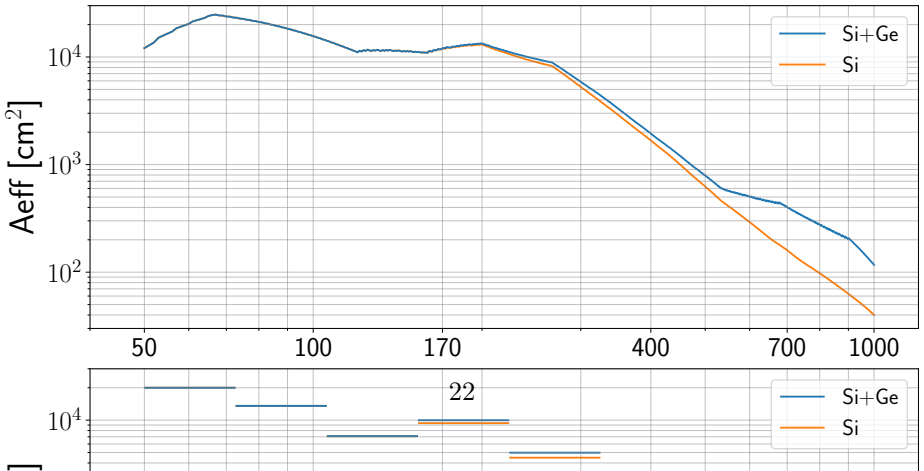


Figure 29



In addition I updated the source generator on MEGAlib taking into account the energy-dependent incident angle of the photons coming from the Laue Lenses for the energy range 50-400 keV. At this stage I only considered the correct geometry for the first order diffractions on the Laue Lenses. For energies 1-50 keV and 400-1000 keV the geometry assumptions were the same as in Figure 7, photon beam hitting the detector azimuthal plane with polar angle $\theta = 0°$ with a diameter of 1.5 mm.

For the Laue Lenses PSF in the energy range 50-400 keV we computed for the PHEMTO focusing distance of 20 m and using only Si(1,1,1) the outgoing photon direction after an incoming photon is diffracted by the Laue Lenses. The Laue lenses optics was assumed to be a planar distribution of individual crystals (no curvature along Z). The Bragg diffraction formula was used to compute at which radial position, r, a given photon energy was emitted. The position on the lenses of the outgoing photons was computed with,

$$r = f\frac{hc}{d_{hkl}E} \tag{20}$$

where f is the focal distance, $hc = 12.4$ AkeV, d_hkl the Si(111) interplane distance and E as the energy of the incoming photon.

The outgoing direction of each photon was computed assuming the photons will focus on the CdTe detector. The Figure below shows, for different incoming photon energies, the diffraction that the Laue Lenses implement on the photons.

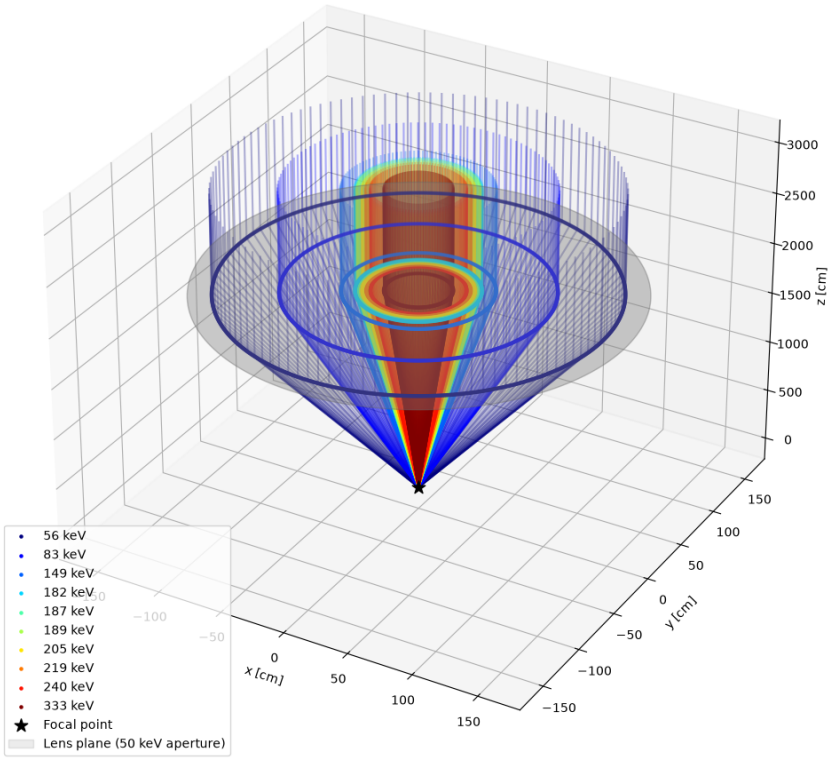


Figure 31

To model the pattern created by an individual energy diffraction on the detection plane we modelled a rectangular

2D source at the position given by 20 with the direction showcased in Figure 31. The dimensions used for this rectangular shape aimed to create a PSF FWHM of 1.5 mm. In the end this information will have to be given by the Italian colleagues but at this stage they said that PSF of 1.5 mm is ok. The emission of photons on this surface was assumed to be isotropic. As an example Figure below shows the case for E = 50 keV for 3 radial positions, $\theta = 0°, 45°, 90°$.

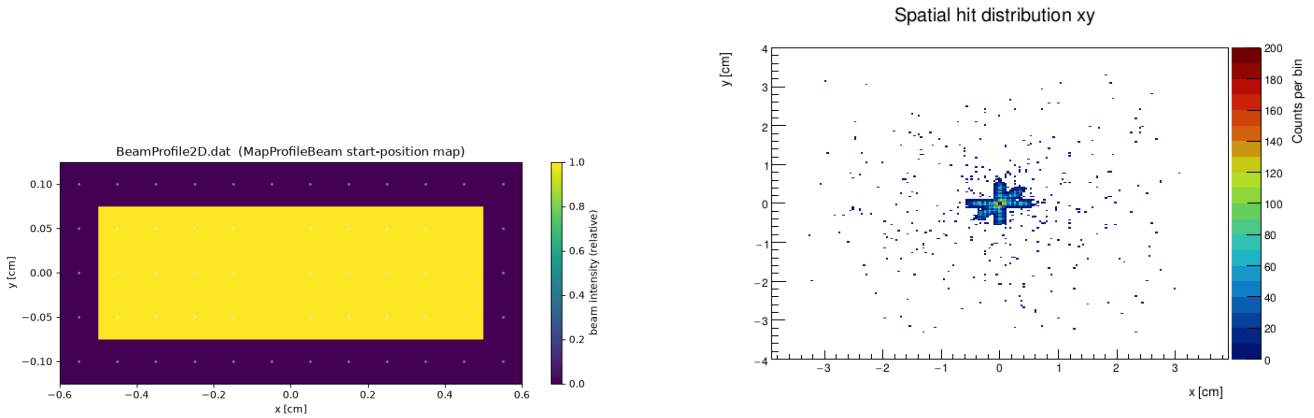


Figure 32: a) Emission pattern from a single drifaction position r. b) Pattern at the focal plane for 50 keV for a Laue Lens positioned at $\theta = 0°, 45°, 90°$

If then one populates all the θ positions for a given radial position (given photon energy), ones gets the following PSF at the focal plane.

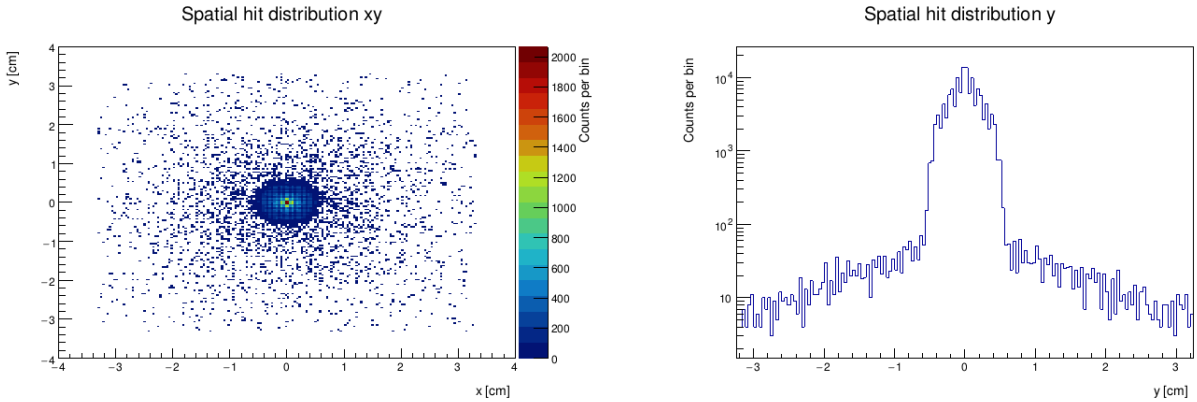


Figure 33: 50 keV

1.6.4 Focal Plane Detection Efficiency

To compute the detector efficiency we used a set of monochromatic sources with the geometry described in the above section *Optics Assumptions*. We use the MEGAlib toolkit to perform the G4 simulations and to perform some pre-processing on the data from the detectors. MEGAlib 2D voxelized the events taking into consideration the pixel dimensions of each detector. In addition it was performed preliminary Compton reconstruction, for example: when a CdTe fluorescence photon exits the CdTe and interacts with the Si, MEGAlib discards this event if the energy distribution is not compatible with the Compton kinematics.

1.6.5 Focal Plane Responce Matrix Funtion - RMF

The RMF was produced with the detector and optics assumptions presented in the section above, namely in the *Focal Plane Detection Efficiency*.

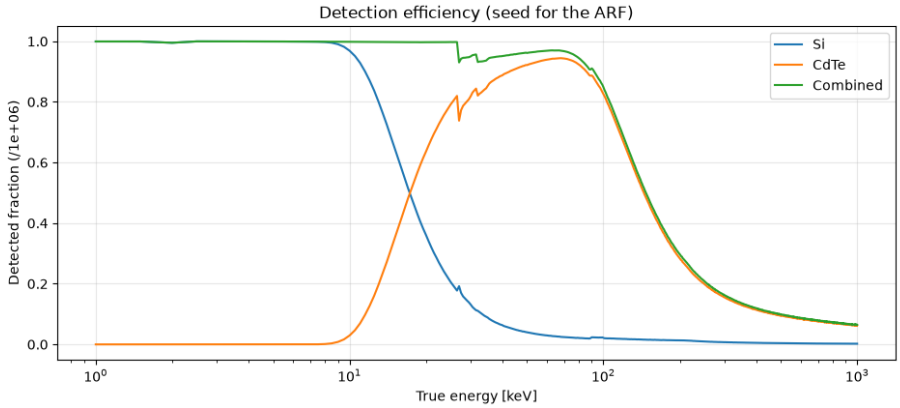


Figure 34

It is also important to note that the definition of an event type in the detector configuration.

- For the LED RMF an event is considered by a time-coincidence of active pixels in the LED detector - This means for example, in the cases where there is fluorescence on the CdTe, the escape photon can interact on the Si and be recorded as an event in the Si.
- For the HED RMF an event is considered by an activation of one or more pixels on the HED detector within a time-coincidence (in reality on G4 output we assigned the same timestamp for the energy deposits from the same parent particle).
- For Both RMF an event is considered as an activation of one or more pixels on either detector within, again, a "time-coincidence". So the total energy of an event, i.e., Compton, can be shared between detectors.

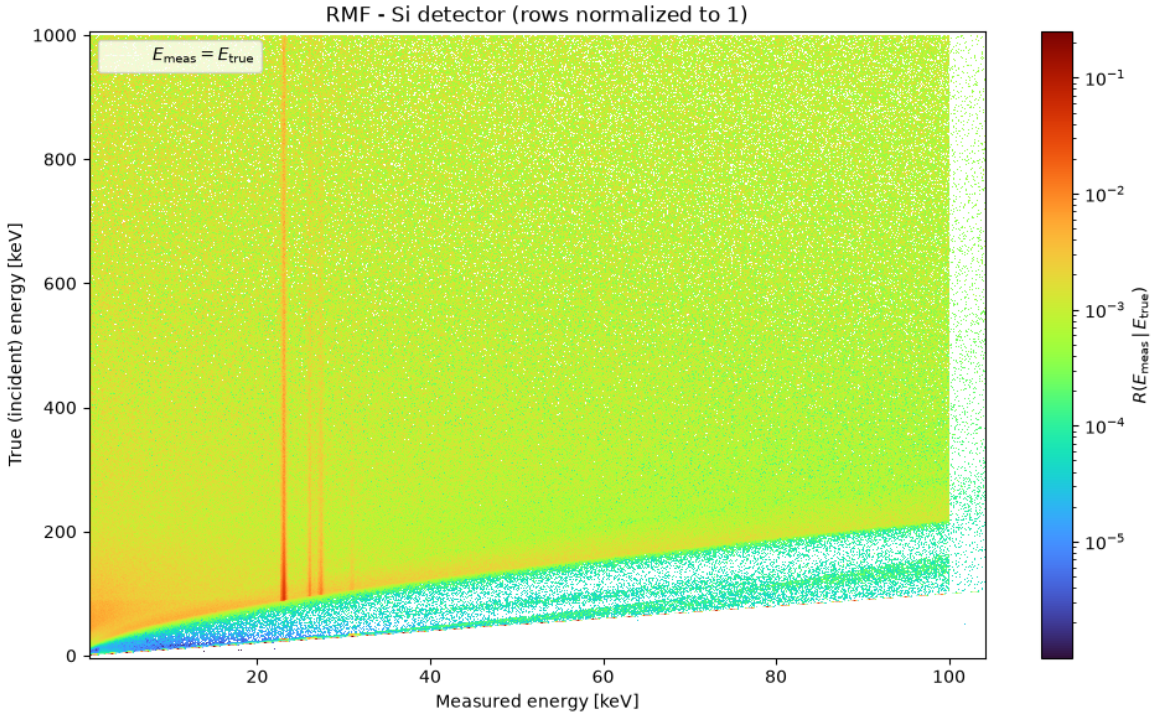


Figure 35: RMF of the LED detector. For this RMF we are just considering events that were uniquely detected by the LED after preliminary Compton acceptance filtering. One can see the vertical structures that are representative of the CdTe fluorescence. Note that below the Compton edge structure MEGAlib removes the fluorescence events because they are Compton incompatible.

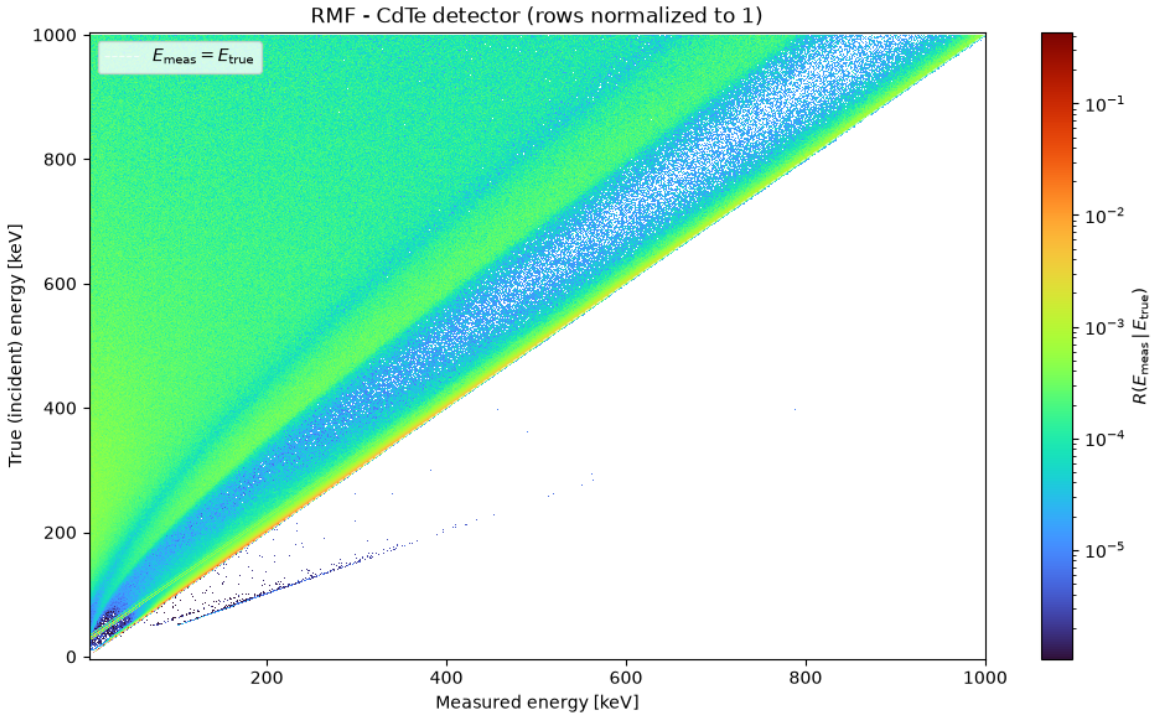


Figure 36: RMF of the HED detector. For this RMF we are just considering events that interacted uniquely in the HED detector, after preliminary Compton acceptance filtering. One can see at lower energies the presence of the Cd and Te K escape lines. As the energy increases the low energy tail explained in the *Detector Assumptions* section above masks the escape lines.

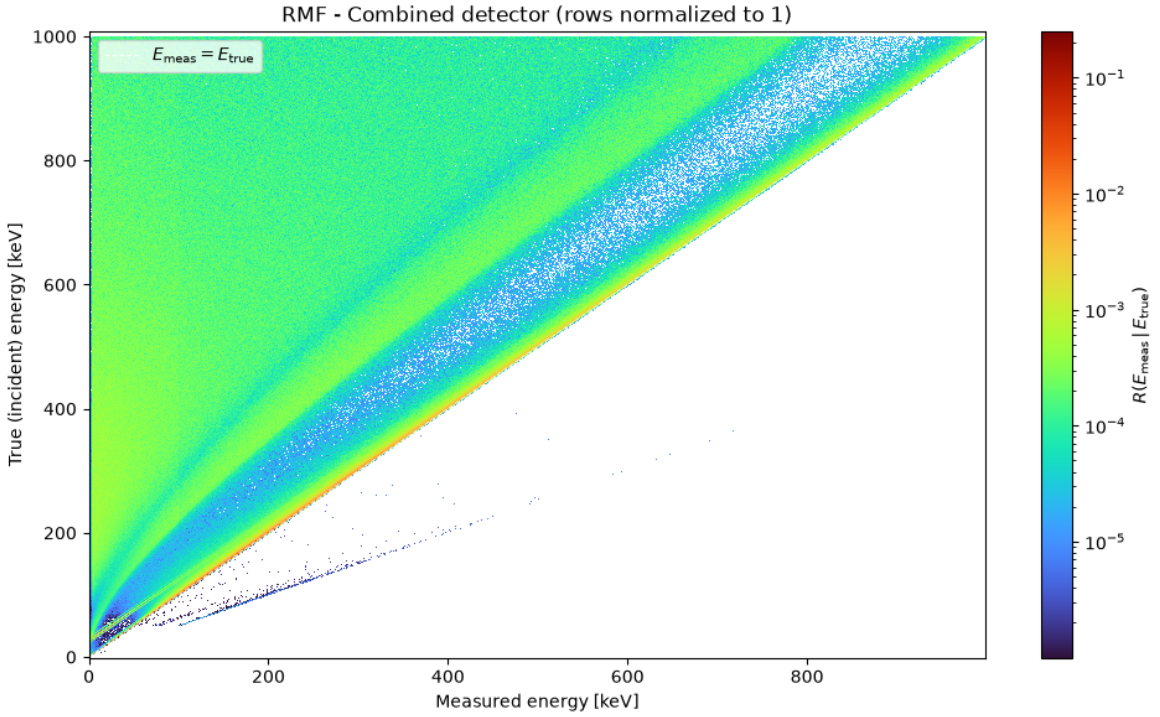


Figure 37: RMF of the Phemto focal plane. For this RMF we are considering all the events in the focal plane. The events can be just from one of the detectors or shared between the two detectors (Compton).

The RMF files are Xpec compatible. Here are some examples for a monoenergetic source.

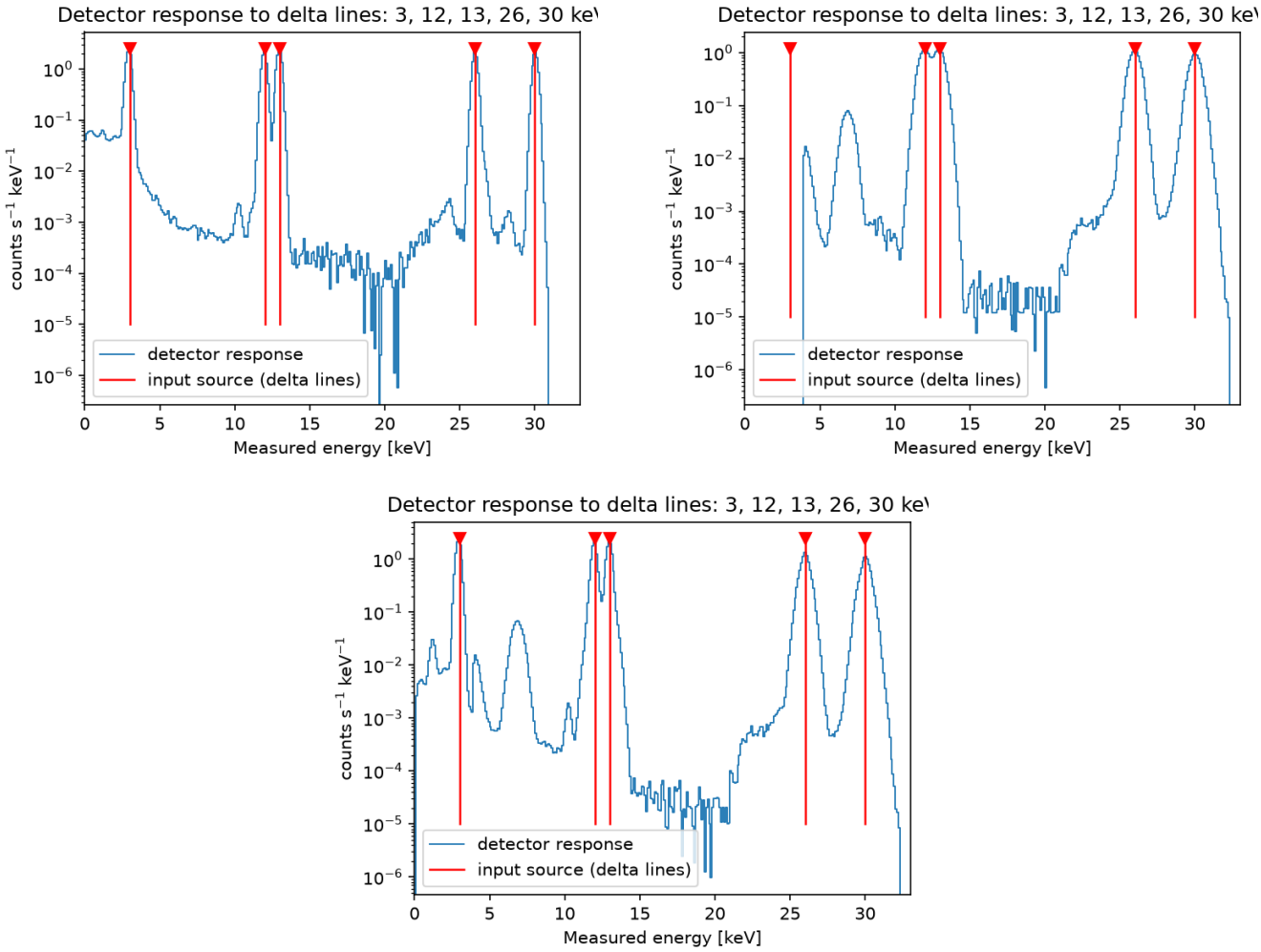


Figure 38: Xspec-compatible RMF examples for a low energy monoenergetic source: LED detector (top left), HED detector (top right) and the combination of the both detectors (bottom).

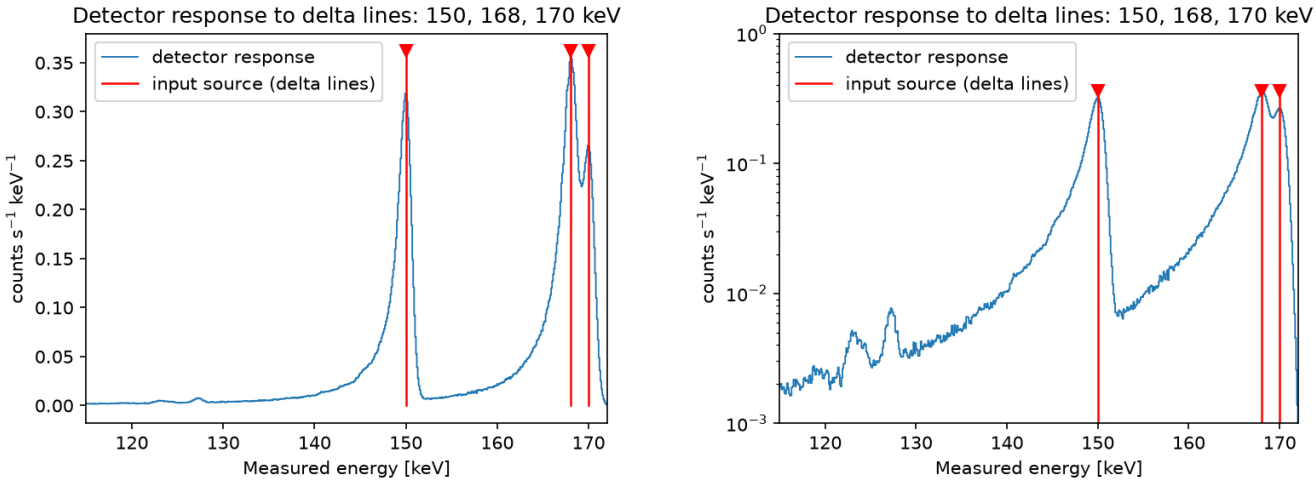


Figure 39: Xspec-compatible RMF examples of the HED for a high energy monoenergetic source: linear scale (left), logarithmic scale (right).

1.6.6 PHEMTO Area Responce Funtion - ARF

PHEMTO is a focusing instrument: two optics concentrate photons onto a small focal spot on the detector: X-ray mirror (1-80 keV) and a Laue Lenses (Si(111) crystals from the most recent updates from the team, check subsection above *Optics Assumptions*, 50-1000 keV). The instrument effective area is the combined optics area times the detector's detection efficiency:

$$A_{eff}(E) = A_{optics} \times \epsilon_{FPA}(E) A_{optics}(E) = A_{mirror} + A_{lenses}(E) \tag{21}$$

The two optics are summed where their bands overlap (50-80 keV). $\epsilon_{FPA}(E)$ is the detection effiency of the focal plane as presented in *Focal Plane Detection Efficiency*.

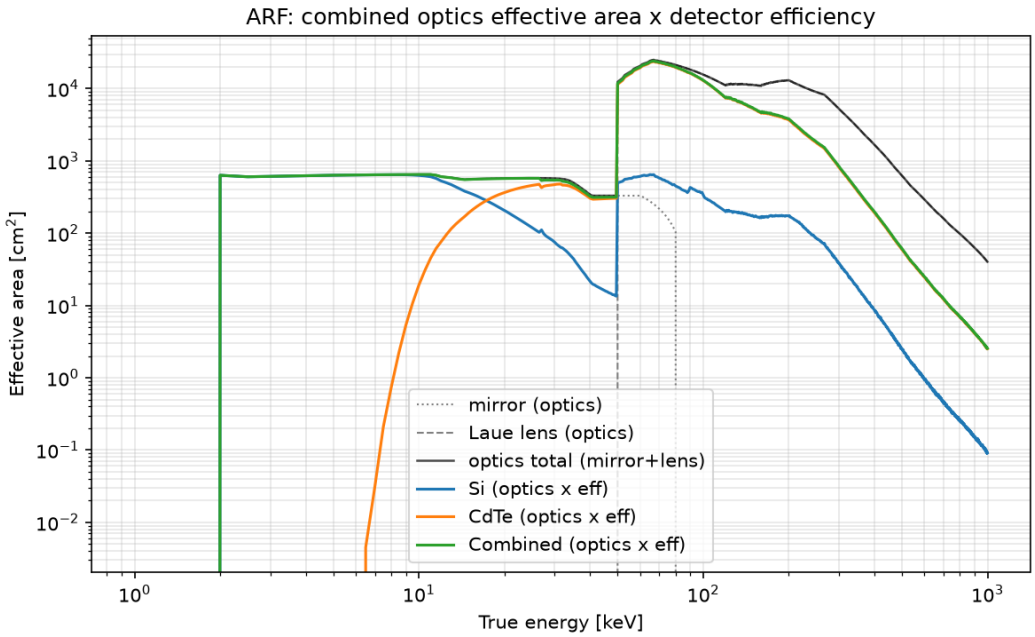


Figure 40

From this we can use Xpec models, for example the Crab.....

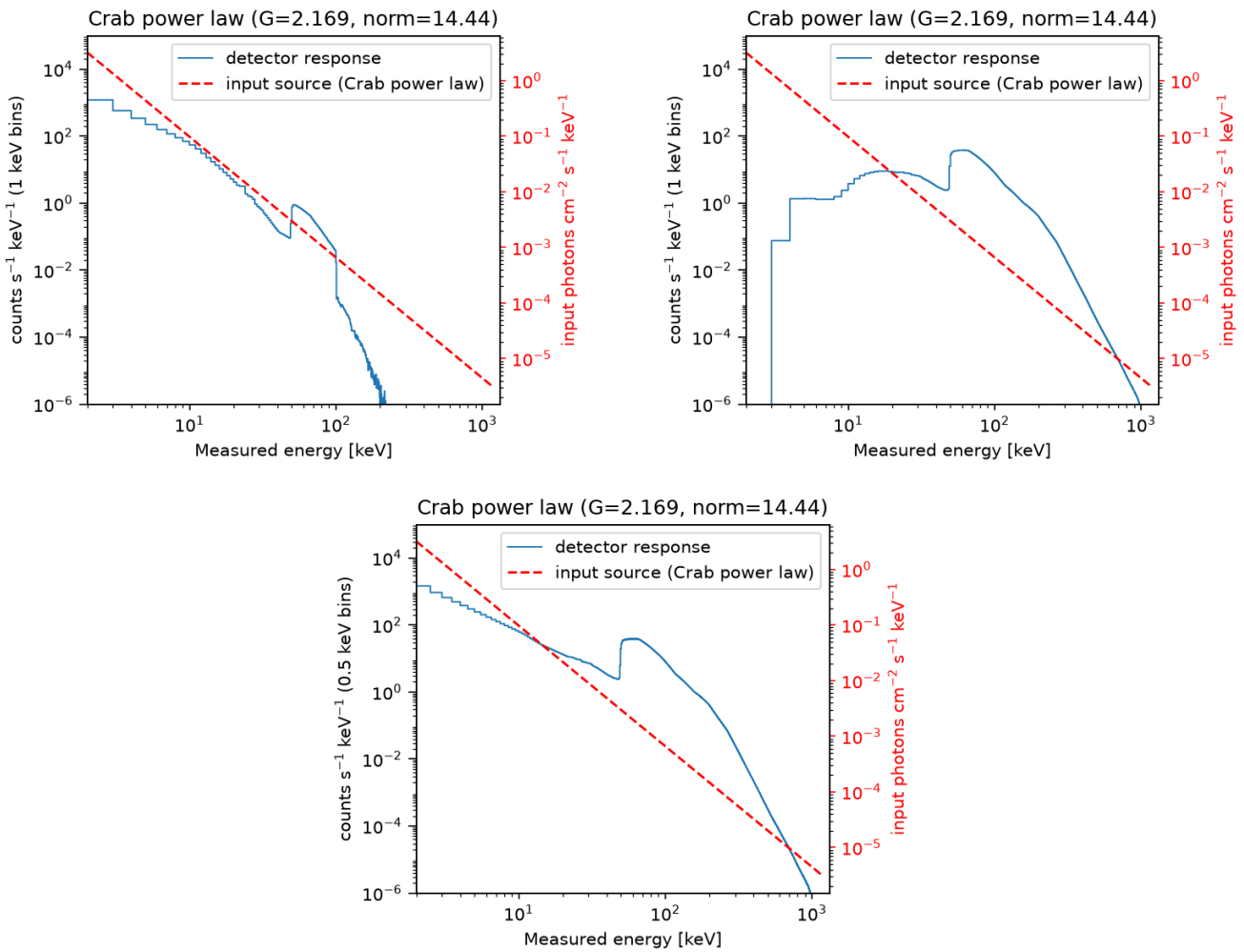


Figure 41: Xspec-compatible RMF examples for a low energy monoenergetic source: LED detector (top left), HED detector (top right) and the combination of the both detectors (bottom).

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